TRIANGULAR PRISM EQUATIONS AND CATEGORIFICATION

ZHENGWEI LIU, SEBASTIEN PALCOUX, YUNXIANG REN, AND GERT VERCLEYEN

Abstract. We obtain the triangular prism equations by evaluating a labelled prism in two ways, as a sum of products of two or of three tetrahedral coefficients. For a spherical fusion category an explicit change of variables identifies these equations with the pentagon equations. Prisms with repeated edge labels single out small systems of coefficients. Using tetrahedral rotations and the Fuchs–Runkel–Schweigert parity theorem, we reduce two such systems to finite solution sets. A prism carrying both distinguished labels excludes every pair of solutions. Two further obstructions, the zero- and one-spectrum criteria, follow directly from the pentagon equations and apply over every field without a pivotal structure. We prove that $\mathcal{F}_{210}$ admits no fusion categorification in characteristic zero and no pivotal fusion categorification in positive characteristic, and that $\mathcal{F}_{660}$ admits no fusion categorification over any field. Together with a bounded census and known unitary obstructions, these results classify, up to Grothendieck equivalence, the non-pointed unitary simple integral fusion categories of rank at most 8 and Frobenius–Perron dimension at most 20000 for which the Frobenius–Perron dimension of each simple object divides that of the category.

1. Introduction

The notion of a *based ring* was introduced by Lusztig [24]. A *fusion ring* is a unital based ring of finite rank; see [11, Definition 3.1.7]. Its categorical counterpart, the *fusion category*, was systematically developed by Etingof, Nikshych and Ostrik [12]. A realization of a fusion ring as the Grothendieck ring of a fusion category is called a *categorification*. Mac Lane's pentagon identity expresses the coherence of the associator [25]. In chosen bases of morphism spaces it becomes the *pentagon equations* (PE) for the F-symbols, generalizations of the $SU(2)$ $6j$-symbols and their quantum analogues [4, 8, 39].

Determining which fusion rings admit categorifications is a central problem in fusion category theory. After choosing bases one seeks invertible associator matrices satisfying the PE, together with unit constraints and evaluation and coevaluation maps satisfying the unit and duality axioms. The individual entries of the matrices are the F-symbols. Solving for them directly is usually difficult. Many constructions of fusion categories therefore proceed by other methods and do not determine their F-symbols explicitly. Finding manageable equations for these coefficients remains useful both for constructing categories and for proving that proposed fusion rules cannot be categorified.

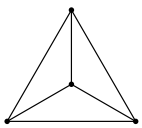
The geometric meaning of the coefficients provides a reason to study them. A spherical fusion category over $\mathbb{C}$ supplies the categorical input for the Turaev–Viro state-sum invariant of 3-manifolds and the associated $(2+1)$-dimensional topological quantum field theory [33, 2]. In a triangulation, edges are labelled by simple objects, triangular faces carry the corresponding morphism data, and tetrahedral weights are expressed through F-symbols. The pentagon relation governs the local change between two and three tetrahedra. In a pointed fusion category, a choice of bases identifies the associator with a group 3-cocycle and the pentagon with its cocycle identity.

The PE have the schematic form

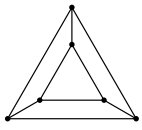
$$\sum_{m} \star\star = \sum_{s} \sum_{m} \star\ \star\ \star, \tag{PE}$$

where each $\star$ denotes an F-symbol, s ranges over simple objects, and m denotes the basis indices in the relevant morphism spaces. The full indexed expression appears in §5.1.

The tetrahedral and triangular prism graphs are the two smallest connected simple planar trivalent graphs:



tetrahedron



triangular prism

In §4.1 and §4.2, we define categorical evaluations of these graphs, with objects on edges and morphisms at vertices. The *triangular prism equations* (TPE) arise by evaluating the same labelled prism as a contraction of two tetrahedra or of three tetrahedra. The graphical notation specifies the ordering of the tensor factors at each vertex and the duality conventions for the edges. Each decomposition determines which coefficients to multiply and which indices to sum over. Thurston's picture [32, Figure 4], reproduced in Figure 1, displays its geometric basis. Categorically,

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the cuts are implemented by dual-basis contraction and the resolution of the identity (§3.1), including the local passage from three parallel strands to two paired trivalent vertices, schematically $||| \mapsto$.

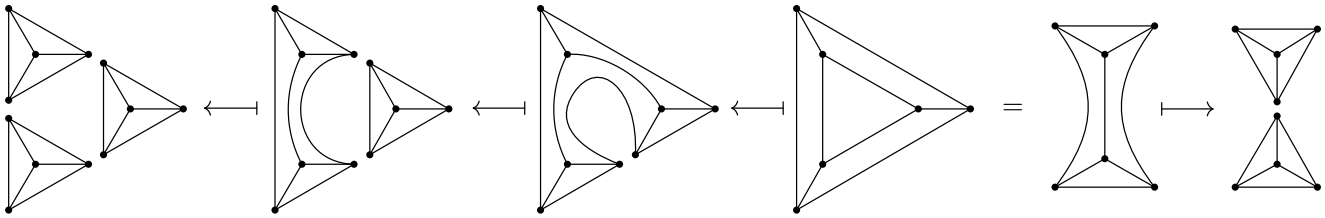


FIGURE 1. Two decompositions of a triangular prism, redrawn from Thurston's picture [32, Figure 4]. Read from the central prism outward: three tetrahedra on the left, two on the right. The intermediate graphs record the cuts underlying the two evaluations.

The geometry also organizes the symmetries. In the spherical setting, rotations of the regular tetrahedron act through A_4, and rotations of the regular triangular prism through the dihedral group of order six, isomorphic to S_3. Under a rotation, each edge keeps its arrow and its object label. Reading the rotated diagram in the original convention replaces a label by its dual when the edge is reversed and cyclically rotates the vertex morphism when its marked corner changes (Proposition 4.17). Consequently, rotationally related decorated tetrahedra give the same scalar, and rotated prism configurations yield corresponding equations. These identifications can reduce both the number of variables and the number of equations that must be retained in a computation.

The change of variables in Theorem 5.3 gives the precise relationship with associativity.

Theorem 1.1. *For a spherical fusion category, the pentagon equations are equivalent to the triangular prism equations.*

In the classical $6j$-symbol formalism, the identity for the move from two tetrahedra to three includes a sum over the new edge, weighted by its categorical dimension [2, §3, Proposition 5.2 and Lemma 5.4]. Theorem 4.12 gives the two cuts of a labelled prism in a pivotal fusion category, with the dual bases, dimension factors and pivotal maps explicit. We look for labellings with few distinct simple objects for which the two cuts still impose a relation on unknown coefficients. For example, we give the six edges of the triangular ends the same label and vary the three connecting labels. The tetrahedra that recur in the two cut diagrams tell us which coefficients to solve for together. A prism carrying two such labels can test whether solutions obtained separately are compatible.

For a finite abelian group G, He, Liu, Lu, Palcoux and Ren derive a variant of Izumi's equations by cutting a prism with the near-group fusion rules $G + |G|$. Over a field in which $|G|$ is invertible, their system characterizes spherical categorifications with $\nu_2(X) = 1$ for every self-dual simple X [16, Theorem 2.3 and Remark 2.4]. Each summand contains two factors of β in their equation (2.7) and three in Izumi's corresponding equation. This form matches the hyperbolic gamma integral they use to construct examples for cyclic G and gives $\beta(2h) = \beta(h)^4$ in characteristic two [16, Remarks 2.2 and 2.6 and §2.2]. Radchenko and Wheeler obtain a corresponding quadratic sum from their finite quantum dilogarithms and use it to derive Izumi's cubic-sum equation [31, Theorem 2 and §5.4].

General categorification criteria. The geometric construction yields two general localization results in §6.2. First, choose a self-dual label k whose tensor square has multiplicity-free, self-dual support S_k, and assume that some coefficient N_{b,b^*}^k is odd. Theorem 6.1 then gives necessary equations for the coefficients for a spherical categorification, using its categorical dimensions. The Fuchs–Runkel–Schweigert theorem supplies the indicator normalization, and tetrahedral symmetry gives, in particular, $y(a, b) = y(b, a)$. Corollary 6.2 restricts exposed labels to a chosen $S'_k \subseteq S_k$ while retaining the full internal summation support.

Second, two such choices $k \neq l$, with $l \in S'_k \cap S'_l$, satisfy the compatibility equation of Theorem 6.4 when their coefficients come from compatible vertex bases in the same category. Separate local systems can therefore have solutions while their coupled system has none. For $\mathcal{F}_{210}$ this gives two systems of twelve equations in ten variables. Each has fourteen complex solutions, and the mixed prism rules out all 196 pairs. This excludes categorification in characteristic zero.

The zero- and one-spectrum criteria (Theorems 6.7 and 6.8) use only the fusion coefficients. The *spectrum* here is the set of simple labels allowed by the fusion rules on the three-factor side of a PE. An empty spectrum or a single summand of multiplicity one can force a contradiction over any field, without pivotality or pseudo-unitarity. The first exclusions for both criteria occur in rank four (Proposition 6.10), and two nonintegral rank-six examples show that neither criterion implies the other (Proposition 6.9). The zero-spectrum criterion excludes $\mathcal{F}_{660}$ over every field. Its proof uses semisimplicity and leaves open realization as the Grothendieck ring of a nonsemisimple finite tensor category. Earlier examples of fusion rings excluded over every field are recorded in [21, §5.2].

The supplementary `GUIDE.md` explains how to apply these criteria to other fusion rings, including localization's dimension input and coupling conventions. The Python interface searches both spectrum criteria and constructs characteristic-zero localization systems with specified or symbolic categorical dimensions. A solution of a necessary subsystem alone does not establish categorifiability.

Applications and motivation. We recall the terminology used in the classification statement. A fusion ring $\mathcal{F}$ is *integral* if the Frobenius–Perron dimensions of its basis elements are integers. It is 1-*Frobenius* if $\mathrm{FPdim}(\mathcal{F})/\mathrm{FPdim}(b)$ is an algebraic integer for every basis element b; in the integral case this means that each $\mathrm{FPdim}(b)$ divides $\mathrm{FPdim}(\mathcal{F})$. This is the usual Frobenius divisibility property. A fusion category has these properties when its Grothendieck ring does. It is *simple* if its only fusion subcategories are itself and the trivial category, and *pointed* if every simple object is invertible. Two fusion categories are *Grothendieck equivalent* when their Grothendieck rings are isomorphic as based rings. We use the standard conventions for fusion rings and fusion categories in [24, 12, 11].

Kaplansky's sixth conjecture [19] asserts that for every finite-dimensional semisimple Hopf algebra H over $\mathbb{C}$, the integral fusion category $\mathrm{Rep}(H)$ is 1-Frobenius. For fusion categories over $\mathbb{C}$, this divisibility is a special case of the strong Frobenius property asked about in [13, Question 1].

Over $\mathbb{C}$, Etingof, Nikshych and Ostrik ask whether every weakly integral fusion category is weakly group-theoretical [13, Question 2]. A non-pointed simple weakly group-theoretical fusion category is equivalent to $\mathrm{Rep}(G)$ for a nonabelian finite simple group G [13, Proposition 9.11]. A simple integral fusion category whose Grothendieck ring is not a group character ring would therefore give a negative answer. More generally, it suffices to find a nontrivial integral fusion category whose Grothendieck ring contains no nontrivial group character fusion subring. Indeed, a minimal nontrivial fusion subcategory of a weakly group-theoretical fusion category is again weakly group-theoretical. In the pointed case its Grothendieck ring is the character ring of a cyclic group of prime order. In the non-pointed case it is the representation category of a nonabelian finite simple group [13, Propositions 4.1 and 9.11]. It is also unknown whether a non-weakly-group-theoretical fusion category can be Grothendieck equivalent to the representation category of a finite group. We therefore seek simple integral fusion rings other than group character rings (or, more generally, integral fusion rings containing no nontrivial group character fusion subring) that pass the known categorification tests.

The fusion rings $\mathcal{F}_{210}$ and $\mathcal{F}_{660}$ arise among the candidates in bounded searches for such examples. The fusion ring $\mathcal{F}_{210}$ is the $q = 6$ member of the interpolation of the character rings of $\mathrm{PSL}(2, q)$ constructed in [22]. This interpolation is a construction of fusion rings from character-table formulas; it does not presume an interpolating family of tensor categories. Its arithmetic properties explain why several familiar necessary conditions for categorification fail to exclude it. Our localization argument rules out any characteristic-zero categorification. A separate dimension-character argument, together with lifting, rules out pivotal categorifications in positive characteristic; see §7.1.

In [23], Wu and two of the present authors applied the Schur product theorem from subfactor theory [20, Theorem 4.1] as a unitary categorification test, the *Schur product criterion*. The primary 3-criterion of [17] refines this approach. Every simple integral fusion ring of rank at most three is pointed (see [10, §1]). Dong, Palcoux and Plessis proved that no simple integral fusion ring of rank four is 1-Frobenius [10, Theorem 1.7]. For ranks five through twelve, the census of 1-Frobenius simple integral fusion rings initiated in [23] gives these bounds [6, §1].

Rank	5	6	7	8	9	10	11	12
FPdim $\leq$	10^7	10^6	10^5	20000	10000	5000	3000	1000.

The census records 505 non-pointed fusion rings. Eight are group character rings, and the primary 3-criterion excludes 477, leaving 20 further candidates. Only two of the latter have rank at most 8, namely $\mathcal{F}_{210}$ and $\mathcal{F}_{660}$. Their unitary categorifiability was asked in [23, Question 1.1]. Our two exclusions answer that question negatively and yield the following conclusion.

Theorem 1.2. *Every non-pointed, unitary,* 1*-Frobenius, simple integral fusion category of rank at most* 8 *with* $\mathrm{FPdim} \leq 20000$ *is Grothendieck equivalent to* $\mathrm{Rep}(\mathrm{PSL}(2, q))$*, where* $q \in \{4, 7, 9, 11\}$.

The four groups are A_5, $\mathrm{PSL}(2, 7)$, A_6 and $\mathrm{PSL}(2, 11)$, of ranks $5, 6, 7, 8$ and dimensions $60, 168, 360, 660$, respectively. The parameter $q = 5$ also gives A_5, whereas $q = 8$ gives rank 9. The theorem concerns Grothendieck equivalence, not uniqueness of the categorification of each surviving fusion ring. Its enumeration input is the bounded census.

The interpolated character rings of [22] also suggest a constructive direction. We hope that the geometric equations and their localized systems will help construct a non-pointed simple integral fusion category outside the representation categories of finite groups. The present obstruction excludes one candidate from that family, while leaving this broader motivation intact. Remark 7.8 discusses the next rank-9 test case and further localizations, together with an alternative approach through induction to the Drinfeld center.

The TPE and small-spectrum criteria have contributed to the classification of multiplicity-free Grothendieck rings [21, §4.1] and the computations through rank seven reported in [36, 34, 37, 35]. Together with the near-group construction discussed above, these applications illustrate that the geometric presentation is useful both for extracting obstructions and for organizing equations in construction problems.

Organization and verification. §§2–3 fix the duality, pivotal and trace conventions. §§4–5 develop the tetrahedral calculus and identify its coefficients with those of the associator. §6 develops localization, and §7 applies the resulting obstructions.

The reader's guide lists the input data and verifiers for each computational claim and distinguishes checking a certificate from generating it. The enumeration of the 505 fusion rings comes from the cited census. The supplement independently checks the 33 entries of rank at most eight and provides the characteristic-zero certificate for $\mathcal{F}_{210}$ and the zero-spectrum check for $\mathcal{F}_{660}$.

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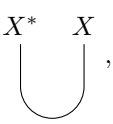
2. Duality and graphical conventions

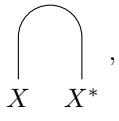
All categories are locally small. We follow the duality conventions of [11, Chapters 2 and 4]. Unless otherwise specified, $\Bbbk$ is an algebraically closed field of arbitrary characteristic. A fusion category over $\Bbbk$ is a $\Bbbk$-linear rigid semisimple monoidal category with finite-dimensional morphism spaces, finitely many isomorphism classes of simple objects, and simple unit $\mathbf{1}$. We allow zero global dimension in positive characteristic. The spectrum criteria in §6.3 apply over an arbitrary field, with $\mathrm{End}_{\mathcal{C}}(\mathbf{1}) = \Bbbk$. We suppress associativity and unit constraints by monoidal coherence. Chosen left duals are denoted by X^*, with

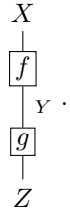
$$\mathrm{ev}_X : X^* \otimes X \longrightarrow \mathbf{1}, \qquad \mathrm{coev}_X : \mathbf{1} \longrightarrow X \otimes X^*.$$

They satisfy the two zigzag identities. We use compatible duals of tensor products, so that $(X \otimes Y)^* = Y^* \otimes X^*$, and normalize $\mathbf{1}^* = \mathbf{1}$. Right duals are denoted by ${}^*X$. Whenever they are used, we suppress the canonical isomorphisms $({}^*X)^* \simeq X$ and ${}^*(X^*) \simeq X$ and their iterates. In particular, an inverse rotation of a last factor X^* uses $X^* \simeq ({}^*X)^{**}$. We write $g \circ f$ for composition, read from right to left; ordinary string diagrams are read from top to bottom.

For $f : X \to Y$ and $g : Y \to Z$, our elementary conventions are

$\mathrm{ev}_X =$ 

, $\mathrm{coev}_X =$ 

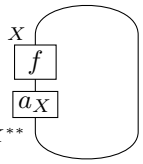
, $g \circ f =$ 

.

The decorated polyhedral graphs of §4.3 will be read as string diagrams using the conventions given there.

2.1. **Pivotal traces and local identities.** A pivotal structure on a rigid monoidal category $\mathcal{C}$ is a monoidal natural isomorphism $a : \mathrm{id}_{\mathcal{C}} \longrightarrow (-)^{**}$. Our trace and dimension conventions are

(2.1)
$$\mathrm{tr}(f) = \quad = \mathrm{ev}_{X^*} \circ ((a_X \circ f) \otimes \mathrm{id}_{X^*}) \circ \mathrm{coev}_X,$$



$$d_X = \dim(X) = \mathrm{tr}(\mathrm{id}_X), \qquad f \in \mathrm{End}_{\mathcal{C}}(X).$$

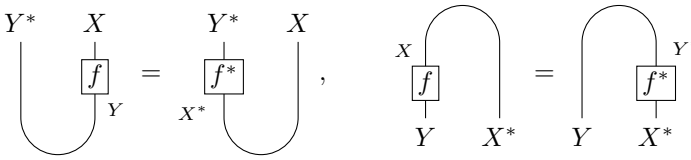
The trace in (2.1) is the left pivotal trace of f. Its right pivotal trace is $\mathrm{tr}(f^*)$, the left pivotal trace of the dual endomorphism $f^* : X^* \to X^*$. A pivotal monoidal category is *spherical* if these traces agree for every f.

Remark 2.1. A pivotal fusion category is spherical if and only if $d_X = d_{X^*}$ for every simple object X [11, Definition 4.7.14 and Theorem 4.7.15]. These are the two pivotal traces of id_X.

Lemma 2.2 (Duality identities)**.** *For $f : X \to Y$, the duality identities are*

$$\mathrm{ev}_Y \circ (\mathrm{id}_{Y^*} \otimes f) = \mathrm{ev}_X \circ (f^* \otimes \mathrm{id}_X), \qquad (f \otimes \mathrm{id}_{X^*}) \circ \mathrm{coev}_X = (\mathrm{id}_Y \otimes f^*) \circ \mathrm{coev}_Y.$$

Graphically, the two identities are



For compatible duality data, $(\mathrm{ev}_X)^* = \mathrm{coev}_{X^*}$ and $(\mathrm{coev}_X)^* = \mathrm{ev}_{X^*}$. For $\alpha : Y \to X^*$ and $\beta : \mathbf{1} \to X$,

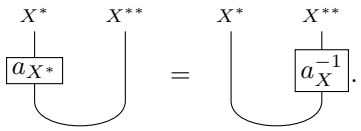
$$\mathrm{ev}_X \circ (\alpha \otimes \beta) = \mathrm{ev}_{X^*} \circ (\beta^{**} \otimes \alpha). \tag{2.2}$$

In a pivotal monoidal category the trace is cyclic, so $\mathrm{tr}(g \circ f) = \mathrm{tr}(f \circ g)$ for $f : X \to Y$, $g : Y \to X$. The following identities also hold, with $f \in \mathrm{End}_{\mathcal{C}}(X)$ in the last formula:

$$\beta^{**} = a_X \circ \beta, \qquad a_X^* = a_{X^*}^{-1}, \qquad a_X^{**} = a_{X^{**}}, \qquad \mathrm{tr}(f^*) = \mathrm{ev}_X \circ (\mathrm{id}_{X^*} \otimes (f \circ a_X^{-1})) \circ \mathrm{coev}_{X^*}. \tag{2.3}$$

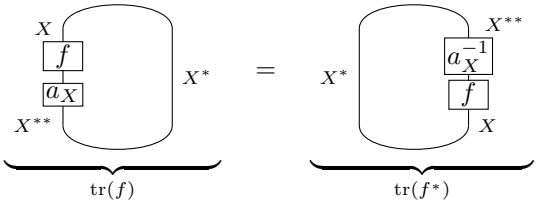
Proof. The first identities are duality naturality and the zigzag identities [11, §2.10]. Both sides of (2.2) equal $\beta^* \circ \alpha : Y \to \mathbf{1}$. The pivotal identities follow from naturality, [11, Exercise 4.7.9], and the trace formulas [11, §4.7]. Sliding f past the cup and cap by the first two identities replaces f by f^{**}, and $f^{**} \circ a_X = a_Y \circ f$ gives cyclicity. □

For example, the pivotal compatibility at a cap is

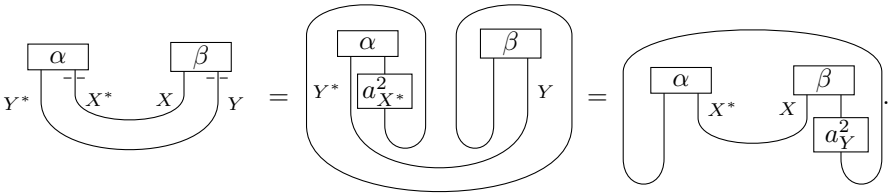


We write $a_X^2 = a_{X^{**}} \circ a_X : X \to X^{****}$.

Lemma 2.3 (Spherical closure)**.** *In a spherical monoidal category, the two closures of $f \in \mathrm{End}_{\mathcal{C}}(X)$ agree:*



For $\alpha : \mathbf{1} \to Y^* \otimes X^*$ and $\beta : \mathbf{1} \to X \otimes Y$, cutting the same closed diagram in different places gives



The dashed cuts on X^* and Y correspond to the middle and last pictures, respectively.

Proof. The first display is sphericality in the conventions of (2.1) and (2.3) [11, Theorem 4.7.15]. Reclosing either marked cut on the opposite side inserts two turns and hence $a_{U^{**}} \circ a_U = a_U^2$ on the indicated leg. □

Lemma 2.4. *For a simple object X in a pivotal fusion category, $d_X \neq 0$ and $f = d_X^{-1}\mathrm{tr}(f)\mathrm{id}_X$ for $f \in \mathrm{End}_{\mathcal{C}}(X)$.*

Proof. The dimension is nonzero by [11, Proposition 4.8.4]. Schur's lemma gives $f = \lambda \mathrm{id}_X$; taking its trace gives $\lambda = d_X^{-1}\mathrm{tr}(f)$. □

2.2. **Oriented cups, caps, and indicators.** In a pivotal monoidal category, write the four duality morphisms as follows [3].

$$\mathrm{ev}_{X,+} = \mathrm{ev}_X, \qquad \mathrm{coev}_{X,+} = \mathrm{coev}_X,$$
$$\mathrm{ev}_{X,-} = \mathrm{ev}_{X^*} \circ (a_X \otimes \mathrm{id}_{X^*}), \qquad \mathrm{coev}_{X,-} = (\mathrm{id}_{X^*} \otimes a_X^{-1}) \circ \mathrm{coev}_{X^*}.$$

The minus-sign evaluation has source $X \otimes X^*$ and the minus-sign coevaluation has target $X^* \otimes X$. Their graphical conventions are

$\mathrm{ev}_{X,+} =$ 

, $\mathrm{ev}_{X,-} =$ 

, $\mathrm{coev}_{X,+} =$ 

, $\mathrm{coev}_{X,-} =$ 

.

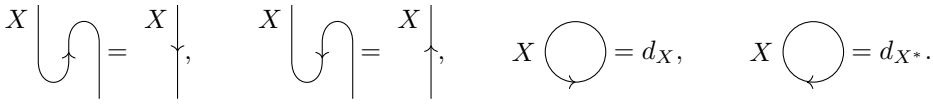
The oriented identity strands are

$\mathrm{id}_X =$ 

, $\mathrm{id}_{X^*} =$

.

The oriented zigzag and loop identities are



Also $(\mathrm{ev}_{X,\pm})^* = \mathrm{coev}_{X^*,\pm}$ and $(\mathrm{coev}_{X,\pm})^* = \mathrm{ev}_{X^*,\pm}$.

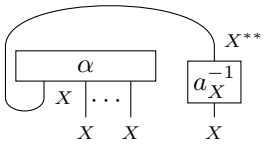
Proof of the oriented identities. Substituting the four definitions reduces the oriented zigzags to the ordinary zigzag identities. The duality formulas follow from Lemma 2.2 and the loop values from (2.1) and (2.3). □

For a simple object X of a pivotal fusion category, with X and X^* fixed, the zigzag identities and the prescribed nonzero pivotal dimensions determine these four morphisms up to the simultaneous rescaling

$$(\mathrm{ev}_{X,+}, \mathrm{ev}_{X,-}, \mathrm{coev}_{X,+}, \mathrm{coev}_{X,-}) \longmapsto (\lambda \mathrm{ev}_{X,+}, \lambda \mathrm{ev}_{X,-}, \lambda^{-1}\mathrm{coev}_{X,+}, \lambda^{-1}\mathrm{coev}_{X,-}), \qquad \lambda \in \Bbbk^*.$$

Proof of the rescaling assertion. Rigidity and simplicity make each of these morphism spaces one-dimensional. Writing the two cap scalings as λ_+, λ_-, the zigzags force the cup scalings to be their inverses. The nonzero loop value then gives $\lambda_+\lambda_-^{-1} = 1$, hence $\lambda_+ = \lambda_-$. □

For $n \geq 1$ in a $\Bbbk$-linear pivotal monoidal category, the cyclic operator on $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X^{\otimes n})$ and the indicator of [29, Definition 3.1, p. 71, and Lemma 3.4, p. 72] are

$E_X^{(n)}(\alpha) =$ 

, $\qquad \nu_n(X) = \mathrm{Tr}\big(E_X^{(n)}\big),$

where Tr denotes the ordinary linear trace whenever this space is finite-dimensional.

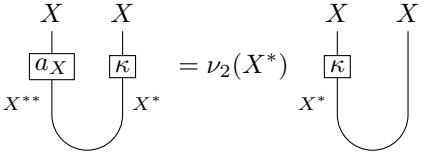
Proposition 2.5. *The operator $E_X^{(n)}$ satisfies $(E_X^{(n)})^n = \mathrm{id}$. If $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X^{\otimes n})$ is one-dimensional, it acts by $\nu_n(X)$ and $\nu_n(X)^n = 1$.*

Proof. The first identity is [29, Theorem 5.1, p. 75]. In a one-dimensional space the operator equals its trace, so the remaining assertions follow. □

Lemma 2.6. *If $\mathrm{Hom}_{\mathcal{C}}(X \otimes X, \mathbf{1})$ is one-dimensional, then every $\kappa : X \to X^*$ satisfies*

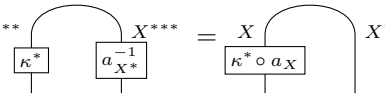
$$\kappa^* \circ a_X = \nu_2(X^*)\kappa, \quad \textit{equivalently} \quad \mathrm{ev}_{X^*} \circ (a_X \otimes \kappa) = \nu_2(X^*)\mathrm{ev}_X \circ (\kappa \otimes \mathrm{id}_X). \tag{2.4}$$

The second identity is the following equality of caps:



.

Proof. Bend κ to $\Phi(\kappa) = (\kappa \otimes \mathrm{id}_{X^*}) \circ \mathrm{coev}_X$. Rotating its first leg gives

$E_{X^*}^{(2)}(\Phi(\kappa)) =$ 

using $(a_X^{-1})^* = a_{X^*}$ at the cup. On this line the rotation acts by $\nu_2(X^*)$; unbending gives (2.4). □

Proposition 2.7. *Let X be a simple object of a pivotal fusion category with $X \simeq X^*$. Then $\nu_2(X)^2 = 1$. If the chosen dual object is literally $X^* = X$, then $a_X = \nu_2(X)\mathrm{id}_X$ and $a_X^2 = \mathrm{id}_X$.*

Proof. Rigidity and simplicity give $\dim_{\Bbbk} \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X \otimes X) = 1$, so Proposition 2.5 gives $\nu_2(X)^2 = 1$. If $X^* = X$, apply Lemma 2.6 with $\kappa = \mathrm{id}_X$. □

Lemma 2.8. *Under the latter hypothesis,*

$$\mathrm{ev}_{X,-} = \nu_2(X)\mathrm{ev}_{X,+}, \qquad \mathrm{coev}_{X,-} = \nu_2(X)\mathrm{coev}_{X,+}.$$

More generally, if $X^{**} = X$ and $a_X = \epsilon \mathrm{id}_X$ with $\epsilon^2 = 1$, then $\mathrm{ev}_{X,-} = \epsilon \mathrm{ev}_{X^*,+}$ and $\mathrm{coev}_{X,-} = \epsilon \mathrm{coev}_{X^*,+}$.

Proof. Substitute $a_X = \nu_2(X)\mathrm{id}_X$, respectively $a_X = \epsilon \mathrm{id}_X$, into the definitions of the four oriented cups and caps; each scalar equals its inverse. □

Literal equality with a chosen dual is not invariant under changing duality data. In particular, in characteristic different from 2, the simultaneous choices $X^* = X$ and $a_X = \mathrm{id}_X$ are incompatible with $\nu_2(X) = -1$. This is a compatibility statement about fixed data, not an obstruction to choosing a self-dual object as its own dual.

3. Pairings, resolutions, and the parity theorem

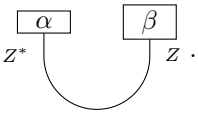
We identify $\mathrm{End}_{\mathcal{C}}(\mathbf{1}) = \Bbbk \mathrm{id}_{\mathbf{1}}$ with $\Bbbk$. We specify the morphism space for each basis and the pairing used to define its dual basis.

3.1. Dual bases and resolution of the identity.

Lemma 3.1. *For any object Z of a fusion category, the pairing*

$$b_Z : \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, Z^*) \times \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, Z) \longrightarrow \Bbbk, \qquad b_Z(\alpha, \beta) = \mathrm{ev}_Z \circ (\alpha \otimes \beta),$$

is nondegenerate. Its graphical form is

$b_Z(\alpha, \beta) =$ 

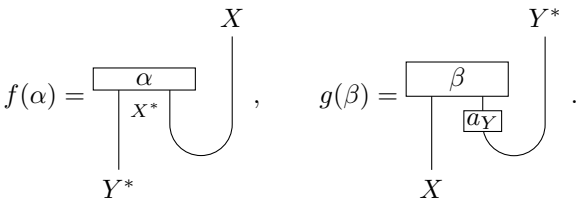
Thus any basis (e_i) of $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, Z)$ has a unique dual basis (e_i') in $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, Z^*)$, and

$$b_Z(\alpha, \beta) = \sum_i b_Z(\alpha, e_i) b_Z(e_i', \beta).$$

Proof. The duality adjunction [11, Proposition 2.10.8] identifies $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, Z^*)$ with $\mathrm{Hom}_{\mathcal{C}}(Z, \mathbf{1})$, under which b_Z becomes composition. This is a perfect pairing on the multiplicity space of $\mathbf{1}$ in the semisimple object Z. □

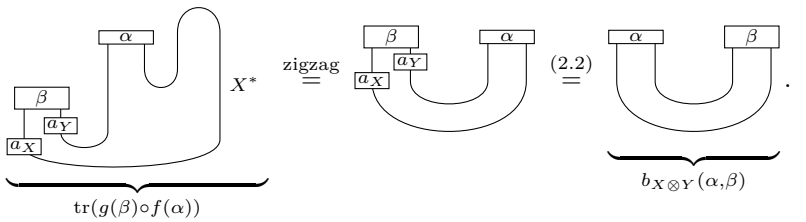
The prime denotes linear duality for this pairing, not the categorical dual $e_i^* : Z^* \to \mathbf{1}$.

Lemma 3.2. *In a pivotal fusion category, define $f : \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, Y^* \otimes X^*) \to \mathrm{Hom}_{\mathcal{C}}(X, Y^*)$ and $g : \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X \otimes Y) \to \mathrm{Hom}_{\mathcal{C}}(Y^*, X)$ by the following bends:*



They are isomorphisms and satisfy $\mathrm{tr}(f(\alpha) \circ g(\beta)) = \mathrm{tr}(g(\beta) \circ f(\alpha)) = b_{X \otimes Y}(\alpha, \beta)$. In particular, if $(u_i), (v_j)$ are dual for $b_{X \otimes Y}$, then $\mathrm{tr}(f(u_i) \circ g(v_j)) = \delta_{ij}$.

Proof. Both bends are duality adjunctions [11, Proposition 2.10.8]. Closing their composite gives

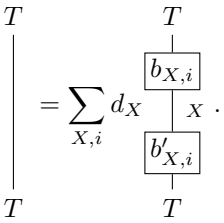


Here $(a_X \otimes a_Y) \circ \beta = \beta^{**}$. Cyclicity gives the other order. □

Proposition 3.3 (Resolution of the identity)**.** *Let T be an object of a pivotal fusion category. For each isomorphism class of simple summands X of T, choose trace-dual bases $b_{X,i} : T \to X$ and $b'_{X,i} : X \to T$, and set $\tau_{X,i,j} = b'_{X,i} \circ b_{X,j}$. Then*

$$\tau_{X,i,j} \circ \tau_{Y,k,\ell} = \delta_{X,Y}\delta_{j,k} d_X^{-1} \tau_{X,i,\ell},$$
$$\gamma = \sum_{X,i,j} d_X \operatorname{tr}(\gamma \circ \tau_{X,j,i}) \tau_{X,i,j} \qquad (\gamma \in \operatorname{End}_{\mathcal{C}}(T)),$$
$$\mathrm{id}_T = \sum_{X,i} d_X \tau_{X,i,i}.$$

The last identity is the graphical resolution



$$= \sum_{X,i} d_X \quad .$$

Proof. Write T as a direct sum of simple objects, with inclusions $q_{X,i}$ and projections $p_{X,i}$. One may take $b_{X,i} = p_{X,i}$ and $b'_{X,i} = d_X^{-1} q_{X,i}$. The elements $d_X \tau_{X,i,j}$ are the ordinary matrix units in the corresponding simple block of $\operatorname{End}_{\mathcal{C}}(T)$. Since $\operatorname{tr}(\tau_{X,i,j}) = \delta_{i,j}$, the trace pairs $\tau_{X,i,j}$ with $d_X \tau_{X,j,i}$. This proves all three formulas, which are unchanged by replacing the bases by any trace-dual pair. □

3.2. **The Fuchs–Runkel–Schweigert parity theorem.** Fuchs, Runkel and Schweigert prove a parity theorem for absolutely simple tensor factors over a field. They observe that no braiding is needed when the factors in reverse order are their respective duals [14, second theorem, p. L257, and Remark (2), p. L259]. The following formulation holds in every characteristic and allows X to be arbitrary.

Theorem 3.4. *Let $\mathcal{C}$ be a pivotal fusion category, X any object, and Y a simple self-dual object. If $\dim_{\Bbbk} \operatorname{Hom}_{\mathcal{C}}(X^* \otimes X, Y)$ is odd, then $\nu_2(Y) = 1$.*

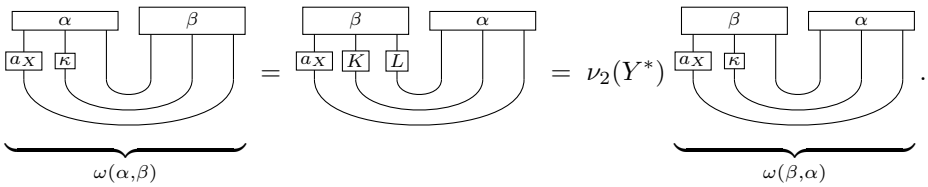
We first prove the symmetry of the relevant pairing without a semisimplicity assumption.

Lemma 3.5 (Parity from a pivotal bilinear form)**.** *Let $\mathcal{C}$ be a $\Bbbk$-linear pivotal monoidal category and suppose that $\operatorname{Hom}_{\mathcal{C}}(Y, Y^*)$ is one-dimensional. Choose a nonzero $\kappa : Y \to Y^*$, put $Z = X \otimes Y \otimes X^*$, and define*

$$f = a_X \otimes \kappa \otimes \mathrm{id}_{X^*} : Z \longrightarrow Z^*, \qquad \omega(\alpha, \beta) = \mathrm{ev}_Z \circ ((f \circ \alpha) \otimes \beta) \quad (\alpha, \beta \in \operatorname{Hom}_{\mathcal{C}}(\mathbf{1}, Z)).$$

Then $\omega(\alpha, \beta) = \nu_2(Y^*)\omega(\beta, \alpha)$ as morphisms in $\operatorname{End}_{\mathcal{C}}(\mathbf{1})$. If moreover $\operatorname{End}_{\mathcal{C}}(\mathbf{1}) = \Bbbk$, the space $\operatorname{Hom}_{\mathcal{C}}(\mathbf{1}, Z)$ is finite-dimensional of odd dimension, and ω is nondegenerate, then $\nu_2(Y^*) = 1$.

Proof. In each coupon the three outgoing legs are X, Y, X^*, from left to right. Write $K = \kappa^* \circ a_Y$ and $L = a_X^* \circ a_{X^*}$. Transpose the nested caps by (2.2) and slide f across:



Indeed $L = \mathrm{id}_{X^*}$ by (2.3), and $K = \nu_2(Y^*)\kappa$ by Lemma 2.6 and adjunction. Since $\nu_2(Y^*)^2 = 1$ by Proposition 2.5, the sign is 1 in characteristic 2. In any other characteristic a nondegenerate skew-symmetric form has even dimension [5, §5, no. 1, Theorem 1], so the sign is 1 in odd dimension. □

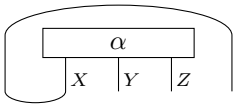
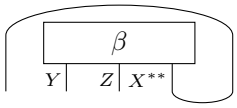
Proof of Theorem 3.4. Choose an isomorphism $\kappa : Y \to Y^*$. Then f is invertible, so Lemma 3.1 makes ω nondegenerate. By rigidity, $\operatorname{Hom}_{\mathcal{C}}(\mathbf{1}, X \otimes Y \otimes X^*) \simeq \operatorname{Hom}_{\mathcal{C}}(X^* \otimes X, Y)$ has odd dimension. Lemma 3.5 gives $\nu_2(Y) = \nu_2(Y^*) = 1$. □

Remark 3.6. Positive multiplicity alone does not imply $\nu_2(Y) = 1$, even for X simple [26, 38]. In Wang's example, $G = \mathrm{PSU}(3, 2) \cong \mathrm{SmallGroup}(72, 41)$ has irreducible complex representations X, Y of degrees $8, 2$, respectively, with $\nu_2(Y) = -1$ and $[X^* \otimes X : Y] = 2$. This is the case $p = 3$ of Miquel Martínez's family [27, Theorem A]. The exhaustive GAP search establishes that 72 is the least possible group order for irreducible X, Y. Its output lists just three further groups through order 128, with identifiers $(128, 764)$, $(128, 801)$ and $(128, 802)$. The first is Mason's example [26]. Among finite simple groups the corresponding search gives $\mathrm{PSU}(3, 5)$, of order 126000, as the first example.

4. Monoidal tetrahedron and triangular prism

4.1. **Monoidal tetrahedron.** Two cyclic permutations of the vertices of a tetrahedron, $(2,3,4)$ and $(3,2,1)$, generate its rotation group A_4. We first describe the corresponding identities for morphisms.

Definition 4.1. Let $\mathcal{C}$ be a monoidal category with left duals. Let F, G and H be functors from $\mathcal{C}^3$ to **Set** given on objects by $F(X,Y,Z) = \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X \otimes Y \otimes Z)$, $G(X,Y,Z) = \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, Y \otimes Z \otimes X^{**})$ and $H(X,Y,Z) = \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X \otimes Y \otimes Z^{**})$, for all objects X, Y, Z in $\mathcal{C}$. Let $\rho : F \to G$ and $\rho' : G \to F$ be natural transformations defined by

$\rho(\alpha) =$  and $\rho'(\beta) =$ 

for all α in $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X \otimes Y \otimes Z)$ and for all β in $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, Y \otimes Z \otimes X^{**})$. Next, assuming the existence of a pivotal structure a, let $\sigma : F \to H$ and $\sigma' : H \to F$ be natural transformations defined by

$$\sigma(\alpha) = (\mathrm{id}_{X\otimes Y} \otimes a_Z) \circ \alpha \quad \text{and} \quad \sigma'(\gamma) = (\mathrm{id}_{X\otimes Y} \otimes a_Z^{-1}) \circ \gamma,$$

for all α in $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X \otimes Y \otimes Z)$ and for all γ in $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X \otimes Y \otimes Z^{**})$.

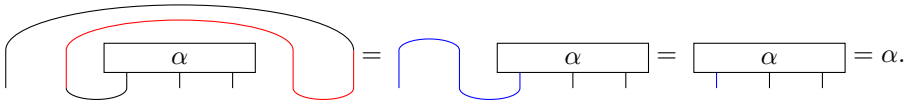
For reference, the first graphical formula is the following typed composite:

$$\rho(\alpha) = (\mathrm{ev}_X \otimes \mathrm{id}_{Y\otimes Z\otimes X^{**}}) \circ (\mathrm{id}_{X^*} \otimes \alpha \otimes \mathrm{id}_{X^{**}}) \circ \mathrm{coev}_{X^*} \in \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, Y \otimes Z \otimes X^{**}). \tag{4.1}$$

Thus a rotation changes the ordered tensor factors; it is not a permutation supplied by a braiding. Iterated symbols such as σ^2 include the appropriate change of object: on the last factor Z they mean $a_{Z^{**}} \circ a_Z : Z \to Z^{****}$, not an untyped square of a_Z.

Lemma 4.2. *Following Definition 4.1, ρ and σ are natural isomorphisms, with $\rho^{-1} = \rho'$ and $\sigma^{-1} = \sigma'$.*

Proof. Put $W = Y \otimes Z$. With the left-dual convention of [11, Proposition 2.10.8(a), p. 42], the two adjunctions identify $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X \otimes W)$ first with $\mathrm{Hom}_{\mathcal{C}}(X^*, W)$ and then with $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, W \otimes X^{**})$. Their composite is ρ; the opposite bends give its inverse. The zigzag identities give

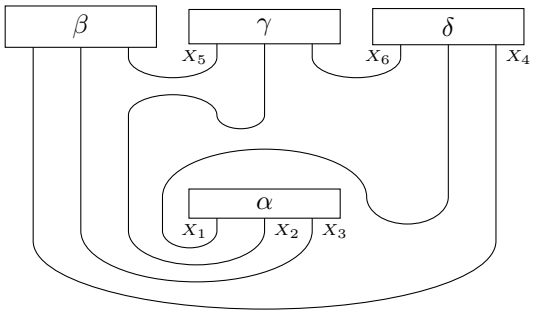
$(\rho' \circ \rho)(\alpha) =$ 

$= \alpha.$

The same identities give $\rho \circ \rho' = \mathrm{id}_G$. Naturality follows from duality naturality, and $\sigma^{-1} = \sigma'$ follows from $a_Z^{-1} \circ a_Z = \mathrm{id}_Z$. □

Remark 4.3. In a $\Bbbk$-linear pivotal monoidal category, the definitions give $(\sigma^{-1} \circ \rho)(\alpha) = E_X^{(3)}(\alpha)$ when $X = Y = Z$. If $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X^{\otimes 3})$ is one-dimensional, Proposition 2.5 thus gives $\rho(\alpha) = \nu_3(X)\sigma(\alpha)$ and $\nu_3(X)^3 = 1$.

Definition 4.4 (Monoidal tetrahedron)**.** Let $\mathcal{C}$ be a monoidal category with left duals. Let $(X_i)_{i=1,\dots,6}$ be objects in $\mathcal{C}$. Consider morphisms $\alpha \in \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X_1 \otimes X_2 \otimes X_3)$, $\beta \in \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X_4^* \otimes X_3^* \otimes X_5^*)$, $\gamma \in \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X_5 \otimes X_2^* \otimes X_6^*)$ and $\delta \in \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X_6 \otimes X_1^* \otimes X_4)$. Let us define the (labeled) *monoidal tetrahedron* as follows:

$T(\alpha,\beta,\gamma,\delta) :=$ 

.

There is also a direct algebraic reading of this diagram. Set $Z = X_1 \otimes X_2 \otimes X_3$ and

$$C_{\beta,\gamma,\delta} = \left(\mathrm{id}_{X_4^*\otimes X_3^*} \otimes \mathrm{ev}_{X_5} \otimes \mathrm{id}_{X_2^*} \otimes \mathrm{ev}_{X_6} \otimes \mathrm{id}_{X_1^*\otimes X_4}\right) \circ (\beta \otimes \gamma \otimes \delta),$$
$$q_\alpha = \mathrm{ev}_Z \circ (\mathrm{id}_{Z^*} \otimes \alpha) : Z^* \longrightarrow \mathbf{1}.$$

Here $C_{\beta,\gamma,\delta} \in \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X_4^* \otimes Z^* \otimes X_4)$, and

$$T(\alpha,\beta,\gamma,\delta) = \mathrm{ev}_{X_4} \circ (\mathrm{id}_{X_4^*} \otimes q_\alpha \otimes \mathrm{id}_{X_4}) \circ C_{\beta,\gamma,\delta} \in \mathrm{End}_{\mathcal{C}}(\mathbf{1}). \tag{4.2}$$

Associativity and unit constraints are suppressed as in §2. Formula (4.2) contracts X_5 and X_6 first, then Z, then X_4; it provides a way to check every source and target without reading a tetrahedral picture in a preferred direction.

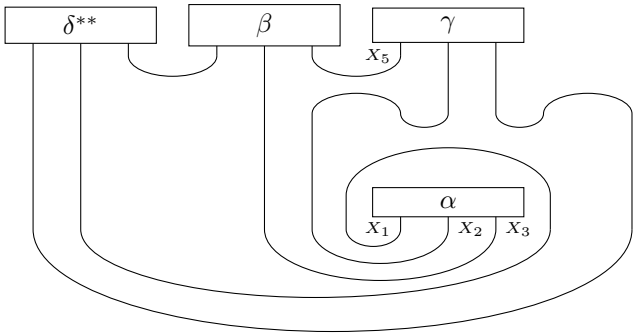
Remark 4.5. From this point on, the unsimplified zigzags in Definition 4.4 (and other figures from this point onwards) are only there to make the figures easier to draw.

Proposition 4.6. *Following Definition 4.4,*

$$T(\alpha,\beta,\gamma,\delta) = T(\rho(\alpha),\delta^{**},\beta,\gamma) = T(\rho^2(\alpha),\gamma^{**},\delta^{**},\beta) = T(\rho^3(\alpha),\beta^{**},\gamma^{**},\delta^{**}) = T(\alpha^{**},\beta^{**},\gamma^{**},\delta^{**}),$$

where ρ is the natural transformation from Definition 4.1.

Proof. Apply the duality identities of Lemma 2.2 to the three legs of δ, with $X = X_6 \otimes X_1^* \otimes X_4$ and $Y = \mathbf{1}$. This gives

$T(\alpha,\beta,\gamma,\delta) =$ 

$= T(\rho(\alpha),\delta^{**},\beta,\gamma).$

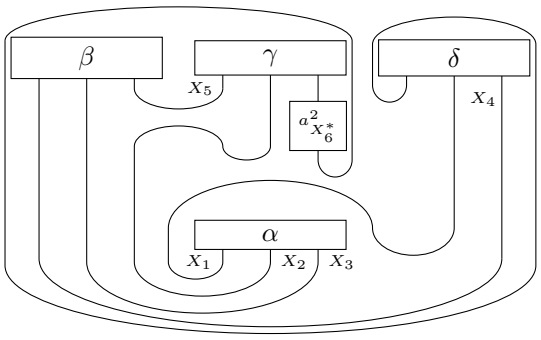
Iterating gives the next two equalities. After three rotations each leg of α has passed once through the duality adjunction, so the resulting morphism is α^{**} under the canonical identification $\mathbf{1}^{**} \simeq \mathbf{1}$. This proves the last equality. □

Proposition 4.7. *If $\mathcal{C}$ is a spherical monoidal category, then*

$$T(\alpha,\beta,\gamma,\delta) = T(\rho^{-3}(\beta),\rho^{-1}(\sigma^2(\gamma)),\rho(\alpha),\rho(\delta))$$

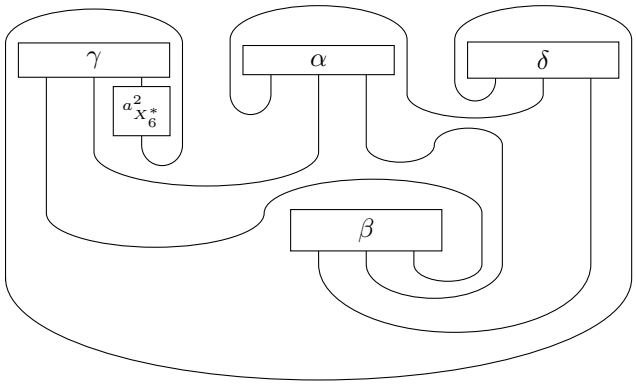
where ρ and σ are the natural transformations from Definition 4.1.

Proof. Cut X_6 and reclose it on the other side of $X_2 \otimes X_3 \otimes X_4$ (Lemma 2.3). This leaves the indicated $a^2_{X_6^*}$ on γ:

$T(\alpha,\beta,\gamma,\delta) =$ 

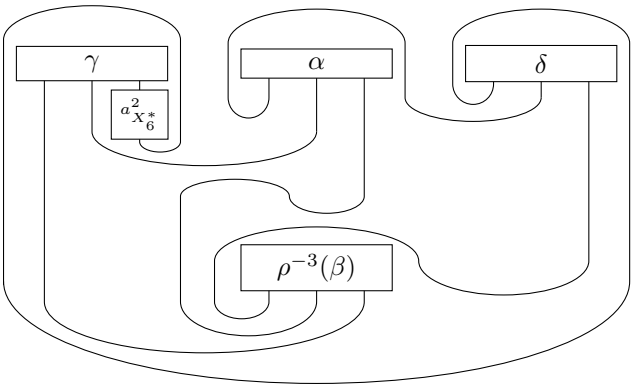
.

Straighten the five unchanged strands:

$T(\alpha,\beta,\gamma,\delta) =$ 

Bend the legs by (4.1); the four corners, in reference order, are $\rho^{-3}(\beta),\ \rho^{-1}(\sigma^2(\gamma)),\ \rho(\alpha),\ \rho(\delta)$:

$T(\alpha,\beta,\gamma,\delta) =$



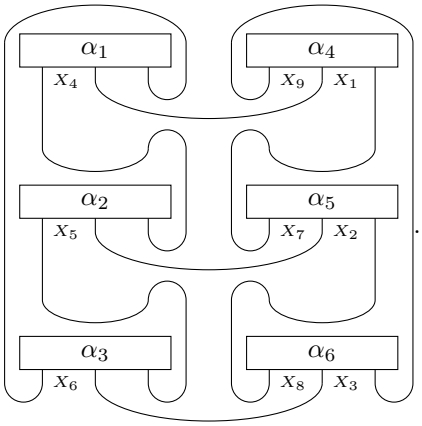
$$= T(\rho^{-3}(\beta), \rho^{-1}(\sigma^2(\gamma)), \rho(\alpha), \rho(\delta)). \qquad \square$$

Remark 4.8. The permutations $(2,3,4)$ and $(3,2,1)$ in Propositions 4.6 and 4.7 generate A_4. Indeed, the subgroup they generate is transitive on the four vertices, lies in A_4, and has at least three elements fixing vertex 1.

4.2. **Monoidal triangular prism and equation.** The triangular prism equation compares two evaluations of the same closed diagram.

Definition 4.9 (Monoidal triangular prism)**.** Let $\mathcal{C}$ be a monoidal category with left duals. Let $(X_i)_{i=1,\dots,9}$ be objects in $\mathcal{C}$. Consider morphisms $\alpha_i \in \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, W_i)$, with $W_1 = X_4 \otimes X_1^* \otimes X_6^{***}$, $W_2 = X_5 \otimes X_2^* \otimes X_4^*$, $W_3 = X_6 \otimes X_3^* \otimes X_5^*$, $W_4 = X_9 \otimes X_1 \otimes X_7^*$, $W_5 = X_7 \otimes X_2 \otimes X_8^*$, $W_6 = X_8 \otimes X_3 \otimes X_9^{***}$. The (labeled) *monoidal triangular prism* is defined as:

$TP(\alpha_1,\alpha_2,\alpha_3,\alpha_4,\alpha_5,\alpha_6) :=$



.

Here is an algebraic reading of the prism. Write $Z = X_1 \otimes X_2 \otimes X_3$. For each vertex $\alpha_i : \mathbf{1} \to P \otimes Q \otimes R^*$ let $f_i = (\mathrm{id}_{P\otimes Q} \otimes \mathrm{ev}_R) \circ (\alpha_i \otimes \mathrm{id}_R) : R \to P \otimes Q$. Along the two triangular faces these maps compose to give

$$F = (f_3 \otimes \mathrm{id}_{X_2^* \otimes X_1^*}) \circ (f_2 \otimes \mathrm{id}_{X_1^*}) \circ f_1 : X_6^{**} \longrightarrow X_6 \otimes Z^*,$$
$$G = (f_4 \otimes \mathrm{id}_{X_2 \otimes X_3}) \circ (f_5 \otimes \mathrm{id}_{X_3}) \circ f_6 : X_9^{**} \longrightarrow X_9 \otimes Z.$$

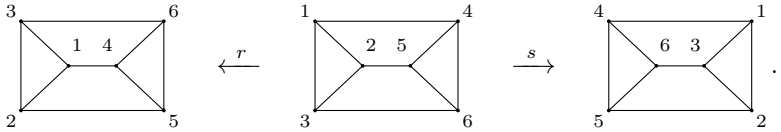
For $h : R^{**} \to R \otimes W$ put $c_R(h) = (\mathrm{ev}_R \otimes \mathrm{id}_W) \circ (\mathrm{id}_{R^*} \otimes h) \circ \mathrm{coev}_{R^*} : \mathbf{1} \to W$. Closing the two faces and evaluating the three joining edges gives

$$TP(\alpha_1,\dots,\alpha_6) = \mathrm{ev}_Z \circ \big(c_{X_6}(F) \otimes c_{X_9}(G)\big) \in \mathrm{End}_{\mathcal{C}}(\mathbf{1}). \tag{4.3}$$

Remark 4.10. Number the vertices of the two triangular faces cyclically as $(1,2,3)$ and $(4,5,6)$, with corresponding vertices joined by edges. The rotations of the regular triangular prism are generated by

$$r = (1\,2\,3)(4\,5\,6), \qquad s = (1\,4)(2\,6)(3\,5).$$

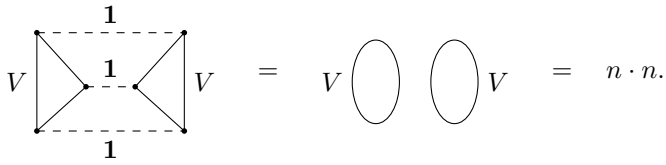
The middle drawing shows the initial numbering. In each adjacent drawing the labels have moved with the indicated rotation.



.

The relations $r^3 = s^2 = 1$ and $srs = r^{-1}$ exhibit the six rotations r^i, sr^i $(0 \le i < 3)$, forming a group isomorphic to S_3. The permutation $(1\,4)(2\,5)(3\,6)$ is the reflection across the plane midway between the triangular faces. In a spherical monoidal category, rotating the graph with its edge orientations, object labels and marked corners preserves its evaluation by spherical isotopy.

Example 4.11 (A complete elementary prism evaluation)**.** In $\mathrm{Vec}_{\mathbb{C}}$, use the canonical identifications with double duals. Let $V = \mathbb{C}^n$, set $X_1 = X_2 = X_3 = \mathbf{1}$ and $X_4 = \cdots = X_9 = V$, and choose all six vertices in Definition 4.9 to be $\mathrm{coev}_V = \sum_a e_a \otimes e^a$ (with the unit factors inserted). The dashed edges represent the unit:



Cutting the three unit edges gives two tetrahedra, each a loop of value n. For the other decomposition the only simple object is $\mathbf{1}$. Identify each vertex space with $V \otimes V^* \simeq \mathrm{End}_{\mathrm{Vec}_{\mathbb{C}}}(V)$. The matrix units E_{ab} and E_{ba} are trace-dual since

$$\mathrm{Tr}(E_{ab} \circ E_{dc}) = \delta_{ac}\delta_{bd}.$$

Each tetrahedron pairs a matrix unit on one cut with the dual basis element on the next, giving

$$\sum_{a,b,c,d,e,f=1}^{n} (\delta_{ac}\delta_{bd})(\delta_{ce}\delta_{df})(\delta_{ea}\delta_{fb}) = \sum_{a,b=1}^{n} 1 = n^2.$$

The simple-object sum has one term, while each vertex-space basis has n^2 elements.

Theorem 4.12 (TPE)**.** *Let $\mathcal{C}$ be a pivotal fusion category. Let a be the pivotal structure. Let $(X_i)_{i=1,\dots,9}$ and $(\alpha_i)_{i=1,\dots,6}$ be objects and morphisms in $\mathcal{C}$ as in Definition 4.9. Then*

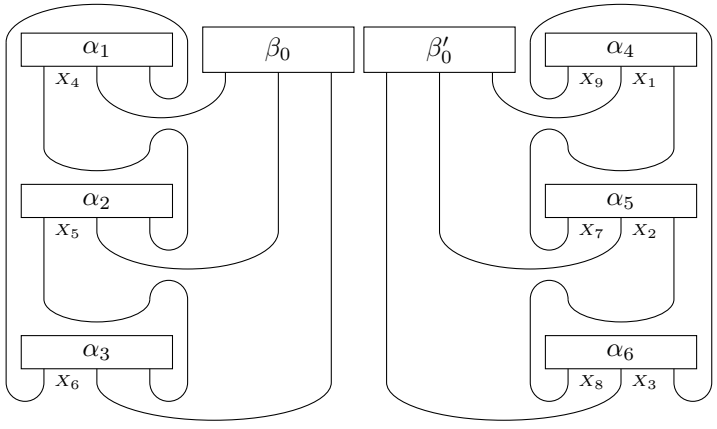
$$\sum_{\beta_0 \in B_0} T\left(\rho^{-2}(\alpha_2), \rho(\alpha_3), \rho^{-1}(\alpha_1), \beta_0\right) T\left(\rho^{-1}(\alpha_5), \beta_0', \rho(\alpha_4), \rho^{-1}(\alpha_6)\right) =$$

$$\sum_{X \in \mathcal{O}} \sum_{\substack{\beta_i \in B_i, \\ i=1,2,3}} \dim(X) T\left(\rho^{-2}(\sigma(\beta_3')), \rho(\alpha_3), \rho^2(\beta_1), \rho^{-1}(\alpha_6)\right) T\left(\rho^{-1}(\alpha_4), \alpha_1, \beta_1', \beta_2\right) T\left(\rho^{-1}(\alpha_5), \alpha_2, \beta_2', \beta_3\right),$$

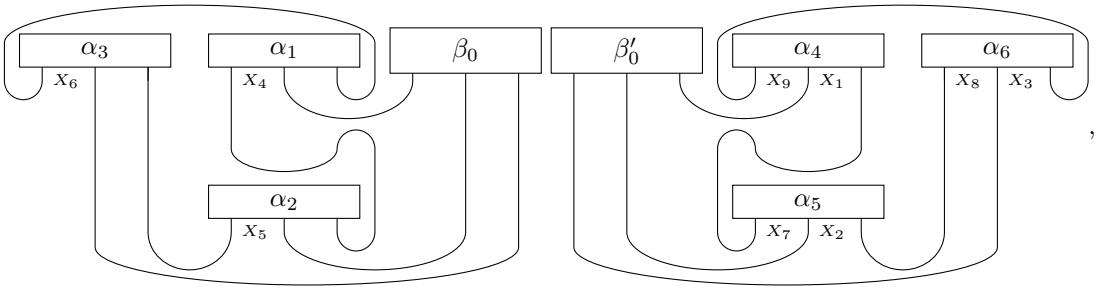
where B_i and B_i' are the bases of $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, Z_i)$ and $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, Z_i^*)$ from Lemma 3.1 ($x \mapsto x'$ denotes the bijection from B_i to B_i'), with $Z_0 := X_1 \otimes X_2 \otimes X_3$, and for $i = 1, 2, 3$, $Z_i := X \otimes Y_i$, where $Y_1 := X_9 \otimes X_6^*$, $Y_2 := X_7 \otimes {}^*X_4$, $Y_3 := X_8 \otimes {}^*X_5$, and $\mathcal{O}$ is the set of simple subobjects of all three objects Y_1^*, Y_2^* and Y_3^* (up to isomorphism). Thus $\mathcal{O}$ contains one representative of each common simple constituent, with no multiplicity. For $i = 1, 2, 3$, the objects and bases in the inner sum depend on X: more explicitly, they are $Z_i(X)$, $B_i(X)$ and $B_i'(X)$. Only Z_0, B_0, B_0' are independent of X. A basis and its dual are chosen separately for each indicated pairing. As before, ρ and σ are the natural transformations from Definition 4.1.

Proof. Write $TP = TP(\alpha_1, \dots, \alpha_6)$.

Step 1: expand the middle cut. On cutting X_1, X_2, X_3, the two sides of the prism give $u : \mathbf{1} \to Z_0^*$ and $v : \mathbf{1} \to Z_0$. Expand $b_{Z_0}(u, v) = \sum_{\beta_0} b_{Z_0}(u, \beta_0) b_{Z_0}(\beta_0', v)$ by Lemma 3.1:

$TP =$ 

Apply the cyclic adjunction (4.1) at the outer vertices and straighten the adjacent cup–cap pairs:

$TP =$ 

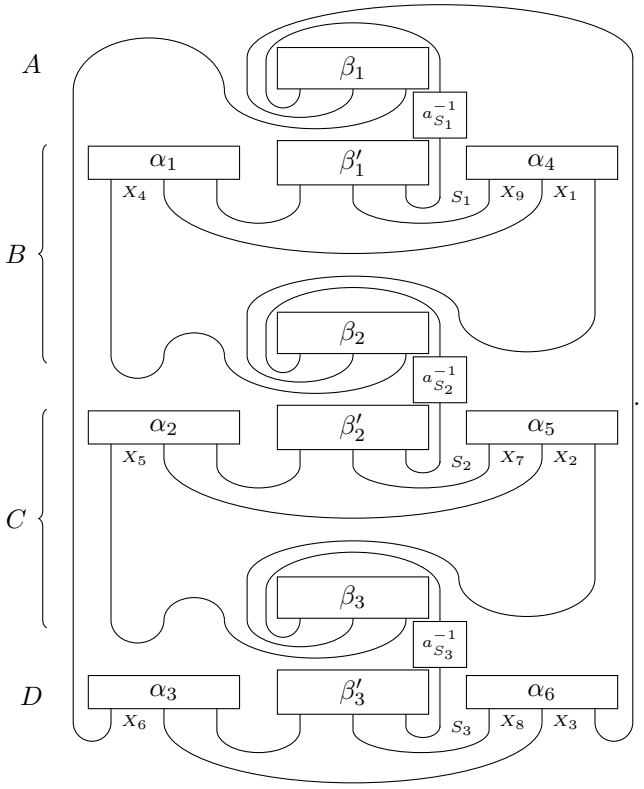
,

The remaining zigzags straighten, giving the two reference tetrahedra:

$$TP = \sum_{\beta_0 \in B_0} T\left(\rho^{-2}(\alpha_2), \rho(\alpha_3), \rho^{-1}(\alpha_1), \beta_0\right) T\left(\rho^{-1}(\alpha_5), \beta_0', \rho(\alpha_4), \rho^{-1}(\alpha_6)\right) = LHS$$

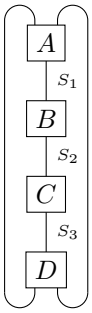
Step 2: resolve the three side cuts. Choose representatives of the simple objects of $\mathcal{C}$ that include those in $\mathcal{O}$. For $T_j = Y_j^*$, let $\mathcal{O}(T_j)$ consist of the representatives occurring in T_j. For each $S_j \in \mathcal{O}(T_j)$, choose a basis $B_j(S_j)$ of $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, S_j \otimes Y_j)$ and its evaluation-dual basis $B_j'(S_j)$ as in Lemma 3.1, using the bases in the statement whenever $S_j \in \mathcal{O}$. Insert the resolution of Proposition 3.3 at each of the three cuts:

$$TP = \sum_{\substack{S_i \in \mathcal{O}(T_i), \\ i=1,2,3}} \sum_{\substack{\beta_i \in B_i(S_i), \\ i=1,2,3}} \prod_{i=1}^{3} \dim(S_i)$$



.

Step 3: close the simple-object blocks. Keep the two outside strands; contract each marked block to its coupon:

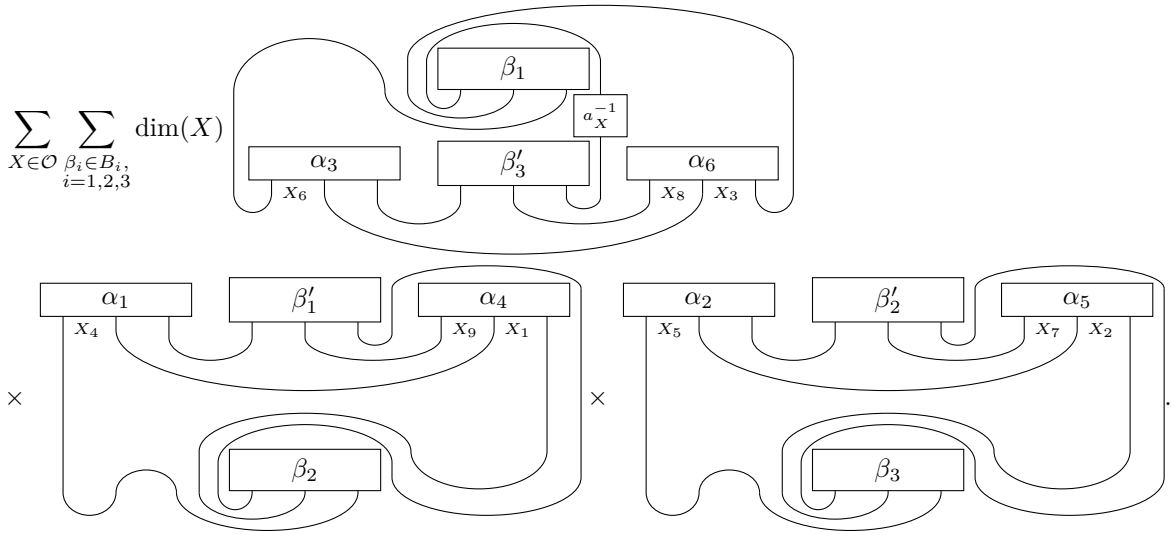


$$= \mathrm{tr}(Q \circ C \circ B) = \frac{\mathrm{tr}(Q)\mathrm{tr}(B)\mathrm{tr}(C)}{\dim(X)^2} \quad (S_1 = S_2 = S_3 = X). \tag{4.4}$$

Here $Q : S_3 \to S_1$ closes A, D along the outside strands. By simplicity, $B : S_1 \to S_2$ and $C : S_2 \to S_3$ give zero unless $S_1 = S_2 = S_3 = X \in \mathcal{O}$. The three bases are then $B_i(X)$, and (4.4) follows from Lemma 2.4. The categorical

dimensions from the three cuts therefore give $\dim(X)^3/\dim(X)^2 = \dim(X)$:

$$TP = \sum_{X\in\mathcal{O}} \sum_{\substack{\beta_i\in B_i,\\ i=1,2,3}} \dim(X)$$

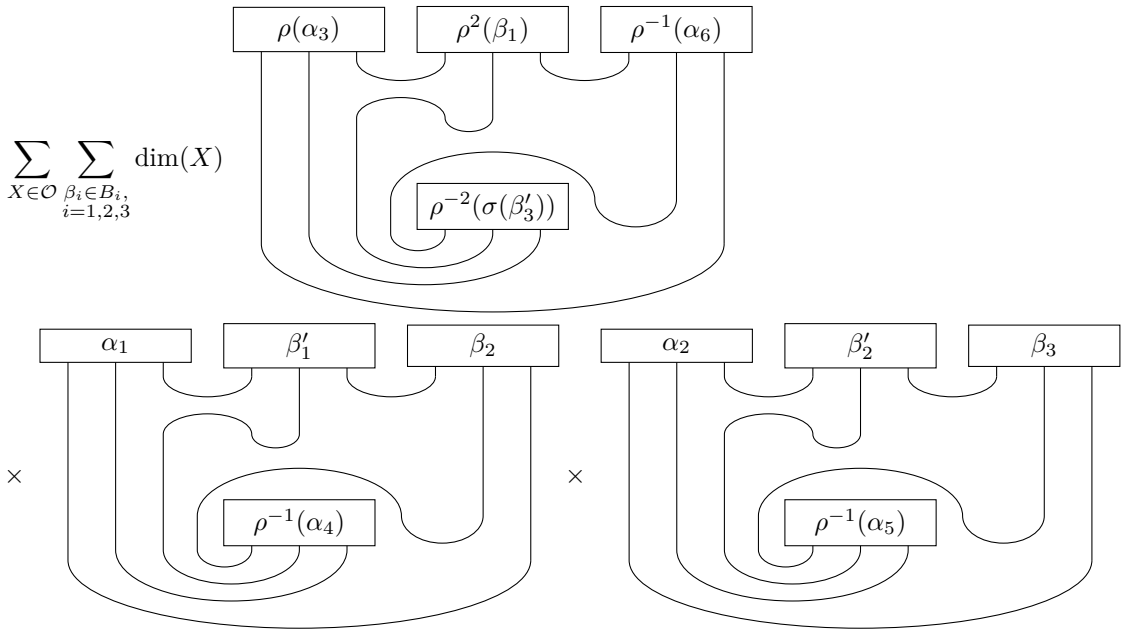


Step 4: convert the three contractions into tetrahedra. Use the zigzag identities for adjacent evaluations and coevaluations, then apply (4.1) (or its inverse) at the vertices. Moving a_X^{-1} across the cup acts on the last factor X^* of β_3' by $(a_X^{-1})^* = a_{X^*}$. Thus

$$\sigma(\beta_3') = (\mathrm{id}_{X_5\otimes X_8^*} \otimes a_{X^*}) \circ \beta_3'.$$

Applying ρ^{-2} gives the first argument of the first tetrahedron. The duality compatibility of Lemma 2.2 then gives the three reference tetrahedra below.

$$TP = \sum_{X\in\mathcal{O}} \sum_{\substack{\beta_i\in B_i,\\ i=1,2,3}} \dim(X)$$



These are precisely the three factors on the right-hand side of the theorem. Combining the two evaluations gives $TP = \mathrm{LHS} = \mathrm{RHS}$. □

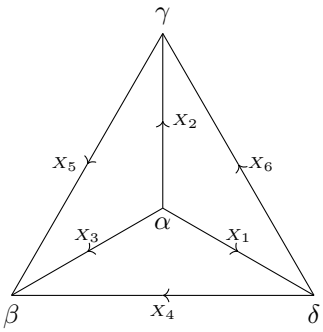
For an explicit type check, call the two tetrahedral factors on the left L_1, L_2 and the three on the right R_1, R_2, R_3, in the order displayed in the theorem. The following table gives the six objects $(U_1, \dots, U_6)$ to substitute for $(X_1, \dots, X_6)$ in Definition 4.4 for each factor. Here ${}^*U$ denotes a right dual, as in the theorem.

	U_1	U_2	U_3	U_4	U_5	U_6
L_1	${}^*X_2$	${}^*X_4$	X_5	X_3	X_6^*	X_1
L_2	${}^*X_8$	X_7	X_2	X_3	X_1	X_9^*
R_1	${}^*X_8$	X^*	X_5	X_3	X_6^*	X_9^*
R_2	${}^*X_7$	X_9	X_1	${}^*X_4$	X_6^{**}	X
R_3	${}^*X_8$	X_7	X_2	${}^*X_5$	X_4	X.

For example, the first vertex of R_1 has type $\rho^{-2}(\sigma(\beta_3')) : \mathbf{1} \to {}^*X_8 \otimes X^* \otimes X_5$, while its third has type $\rho^2(\beta_1) : \mathbf{1} \to X_6^* \otimes X^{**} \otimes X_9^{**}$. These are precisely $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, U_1 \otimes U_2 \otimes U_3)$ and $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, U_5 \otimes U_2^* \otimes U_6^*)$. The other entries are checked in the same way by the explicitly typed rotation (4.1).

4.3. **Oriented monoidal tetrahedron and spherical TPE.** With the oriented duality conventions of §2.2, the action of A_4 on a tetrahedron determines how to change each vertex corner and edge label. We use these rules to express the TPE in the spherical graphical calculus [2, §3].

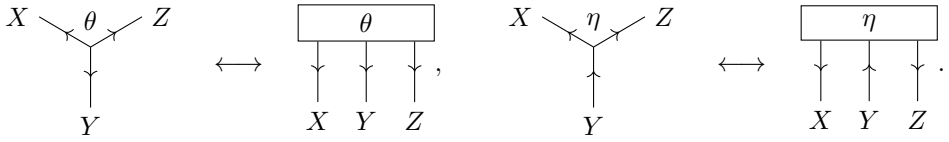
Definition 4.13. Following Definition 4.4,



$:= T(\alpha, \beta, \gamma, \delta).$

The graph denotes the scalar $T(\alpha, \beta, \gamma, \delta)$ of Definition 4.4; the drawing in that definition gives its top-to-bottom reading as a string diagram.

The marked corner determines the ordered tensor factors. With that corner at the top, the local dictionary is



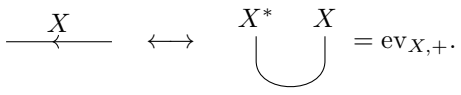
Here $\theta : \mathbf{1} \to X \otimes Y \otimes Z$ and $\eta : \mathbf{1} \to X \otimes Y^* \otimes Z$. The whole marked vertex may be rotated to this position; its corner must not be moved independently. Object labels may be placed anywhere along their edges.

The following dictionary fixes the distinction between the two kinds of diagrams.

Notation	Categorical meaning
Box with ordered legs	A morphism with a specified source and target, read with the convention of §2.
Vertex with a marked corner	A morphism $\theta : \mathbf{1} \to U \otimes V \otimes W$; the corner determines the first factor, hence the ordered triple.
Arrow on an edge	Specifies which endpoint contributes X and which contributes X^* to the evaluation. Reversing the arrow also changes its object label to a dual and changes the adjacent corner data according to (4.5)–(4.6).
$\rho(\theta)$	The cyclic adjunction (4.1), with target $V \otimes W \otimes U^{**}$.
$\sigma(\theta)$	Composition with $\mathrm{id}_{U\otimes V} \otimes a_W$, with target $U \otimes V \otimes W^{**}$.
θ'	The basis element dual to θ under the evaluation pairing of Lemma 3.1. The categorical dual is the morphism θ^*.
Closed tetrahedral graph	The scalar (4.2), after conversion to the reference pattern.

We introduce two rules, (4.5) and (4.6). Each changes an edge orientation together with a dual label and an adjacent vertex corner. They are not independent permissions to reverse an arrow or to move a label while leaving the other data fixed. The rule is applied at the indicated vertex. At the opposite endpoint, the reversed, dual-labelled edge represents the same tensor factor, so the morphism and its marked corner remain unchanged. In Definition 4.13, the tetrahedral graph, together with its prescribed edge orientations and choice of vertex corners, is taken as the *reference pattern*. Other decorated tetrahedra are interpreted using these corner rules, the spherical symmetries below, and the oriented cups and caps of §2.2, retaining every pivotal morphism introduced by the conversion to the reference pattern.

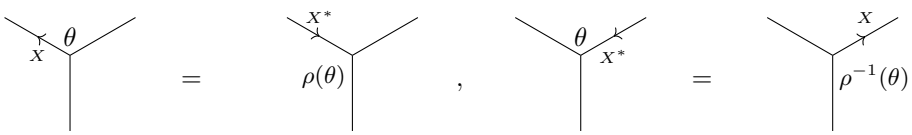
The indicated orientation represents the evaluation morphism



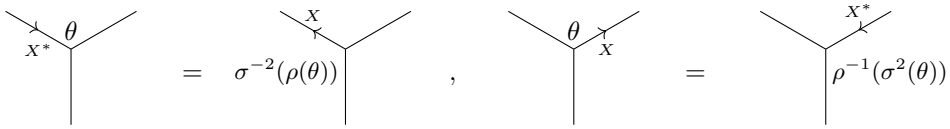
$= \mathrm{ev}_{X,+}.$

The corresponding cup–cap cancellations are the zigzag pictures of §2.2. In the following local rules an unarrowed leg means that both choices of orientation are permitted; it is not the orientation-free specialization of Remark 4.25.

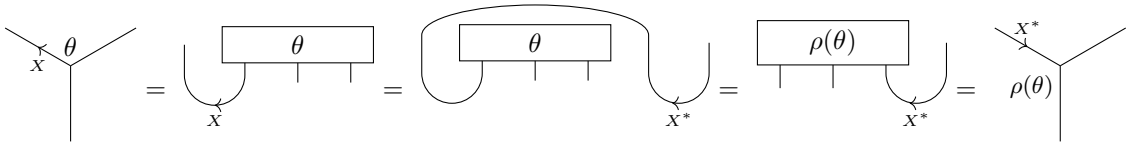
(4.5)



(4.6)



Proof of the local rules. The first rule of (4.5) is the defining duality adjunction for ρ:



Reversing this calculation gives the second rule. The same adjunction with X replaced by X^* gives (4.6): the pivotal-dual identities of Lemma 2.2 identify the resulting double-dual legs, with $a_{X^{**}} \circ a_X$ or its inverse on the affected leg. These are precisely σ^2 and σ^{-2} in (4.6). □

Example 4.14 (Reading and changing a decorated tetrahedron)**.** Use a strictly pivotal realization of a spherical monoidal category (Remark 4.22), so that $(-)^{**} = \mathrm{Id}$ and $a_U = \mathrm{id}_U$. Write $H(U,V,W) = \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, U \otimes V \otimes W)$.

(a) Read the reference graph. Use the graph of Definition 4.13, with the six objects $X_1, \dots, X_6$ and labels

label	Hom-space
α	$H(X_1, X_2, X_3)$
β	$H(X_4^*, X_3^*, X_5^*)$
γ	$H(X_5, X_2^*, X_6^*)$
δ	$H(X_6, X_1^*, X_4)$.

The evaluation is $T(\alpha, \beta, \gamma, \delta)$ from (4.2).

(b) Reverse one edge by (4.5). Reverse the X_5 edge at γ, replacing its label by X_5^*. The first graph below has

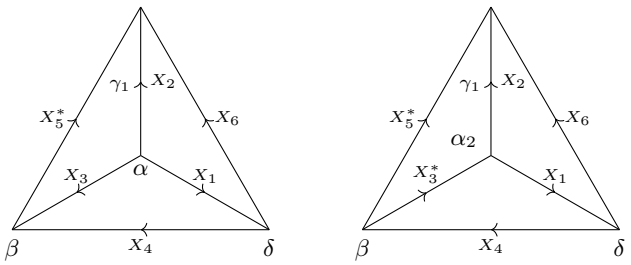
$$\gamma_1 = \rho(\gamma) \in H(X_2^*, X_6^*, X_5).$$

At β, the corresponding tensor factor is still X_5^*.

(c) Move another corner by (4.6). Reverse the last edge X_3 at α and label it X_3^*. Since $\sigma = \mathrm{id}$, the central vertex of the second graph is

$$\alpha_2 = \rho^{-1}(\alpha) \in H(X_3, X_1, X_2).$$

With a general pivotal structure this label would be $\rho^{-1}(\sigma^2(\alpha))$, as prescribed by (4.6).



(d) Translate back. Returning both corners to the reference pattern gives

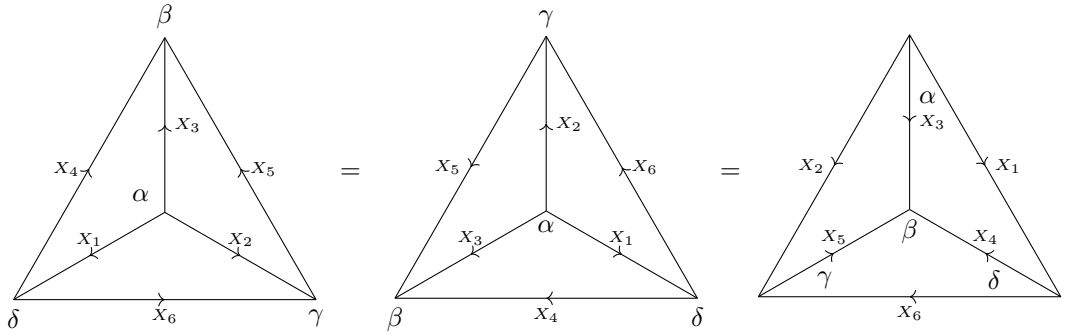
$$T\big(\rho(\alpha_2), \beta, \rho^{-1}(\gamma_1), \delta\big) = T(\alpha, \beta, \gamma, \delta). \tag{4.7}$$

For finite-dimensional vector spaces, choose bases for the six edge objects. In these coordinates

$$T(\alpha, \beta, \gamma, \delta) = \sum_{a,b,c,d,e,f} \alpha_{abc} \beta_{dce} \gamma_{ebf} \delta_{fad}.$$

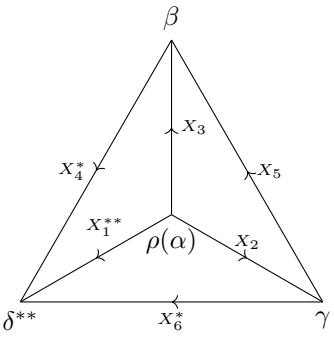
The new coefficients $(\alpha_2)_{cab} = \alpha_{abc}$ and $(\gamma_1)_{bfe} = \gamma_{ebf}$ give the same sum.

Lemma 4.15. *Following Definition 4.13 and above rules in the spherical case*



Proof. By Proposition 4.6, $T(\alpha,\beta,\gamma,\delta) = T(\rho(\alpha),\delta^{**},\beta,\gamma)$, but by Definition 4.13:

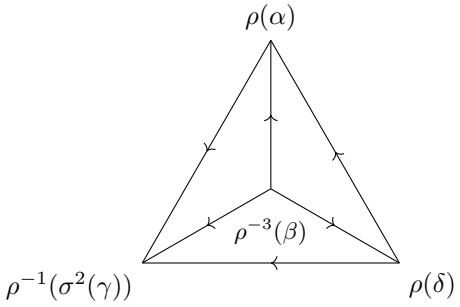
$T(\rho(\alpha),\delta^{**},\beta,\gamma) =$



Now $\delta^{**} = \rho^3(\delta)$, so three applications of (4.5) at its vertex and one at $\rho(\alpha)$ give the first equality. For the second, Proposition 4.7 gives

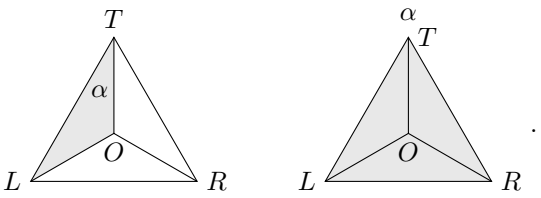
$$T(\alpha,\beta,\gamma,\delta) = T\big(\rho^{-3}(\beta),\rho^{-1}(\sigma^2(\gamma)),\rho(\alpha),\rho(\delta)\big).$$

The reference graph of the right-hand side is



Here the edge labels are determined by the displayed morphisms and are suppressed. Apply (4.5) once at $\rho(\alpha)$, three times at $\rho^{-3}(\beta)$, then (4.6) once at $\rho^{-1}(\sigma^2(\gamma))$, and (4.5) once at $\rho(\delta)$. This gives the last graph in the statement. □

Remark 4.16. Observe that the three pictures in Lemma 4.15 use exactly the same labels. What is changing is the ordering, the orientation of some edges and the corner of the label of some vertices. The twelve orientation-preserving rotations of the regular tetrahedron give equal evaluations, with all corner and orientation data transported. They form the A_4-orbit of the marked reference tetrahedron; coincident decorations can make distinct rotations yield the same picture. The two equalities of Lemma 4.15 correspond to two generators of the group A_4 (see Remark 4.8). A marked corner is equivalently a marked incident face of the spatial tetrahedron. The shaded examples show an interior corner and the exterior corner at T:



face TLO hidden face TLR

Thus a spatial rotation transports a vertex and its chosen incident face together, including the face hidden by the planar projection.

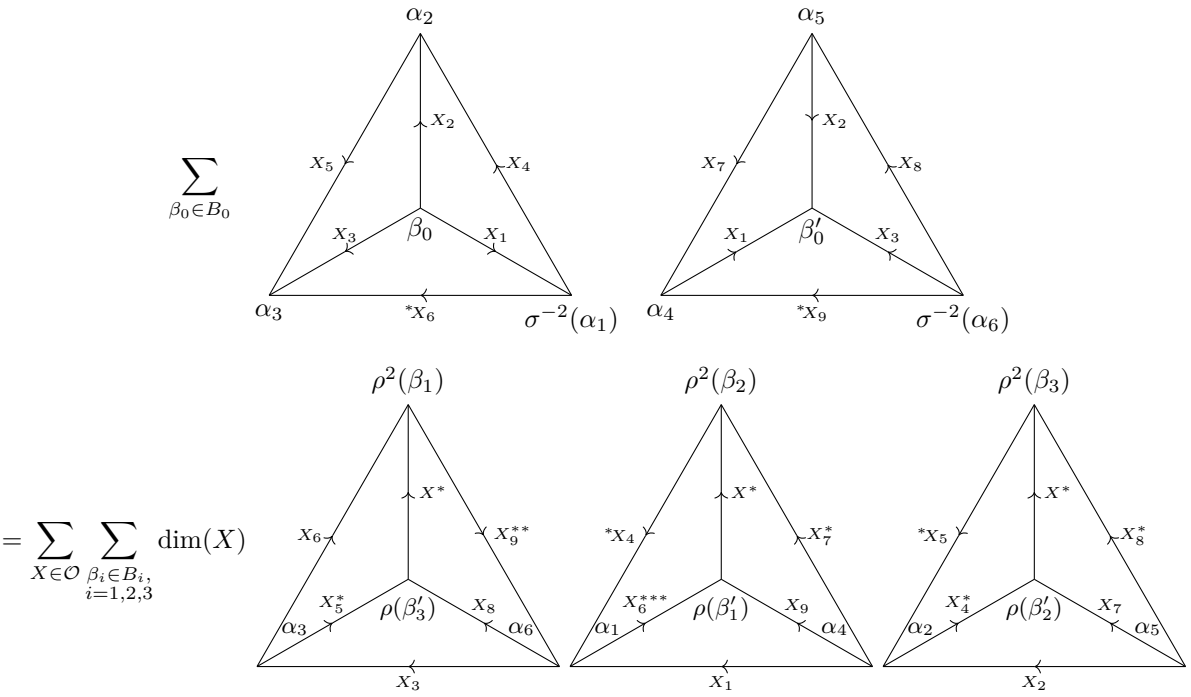
Let us restate this into the following proposition:

Proposition 4.17. *In a spherical monoidal category, let the alternating group A_4 act on the set of labeled oriented tetrahedral graphs as described in Remark 4.16, and consider the orbit of the graph in Definition 4.13. Any element of this orbit can be transformed into the reference pattern of Definition 4.13 by applying the rules above one or more times. Moreover, its categorical interpretation agrees with that of the original graph.*

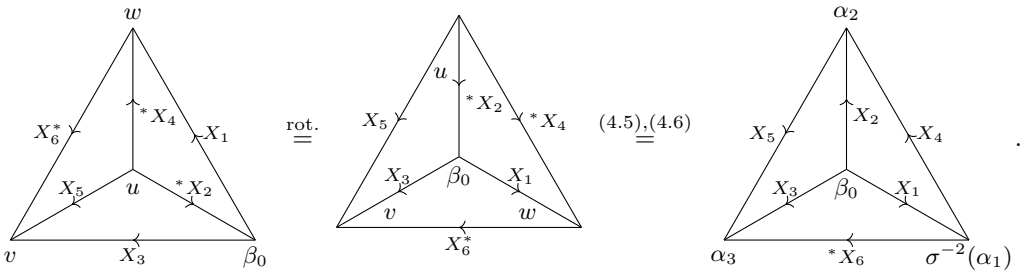
Proof. The two graphical equalities of Lemma 4.15 realize the permutations (2 3 4) and (3 2 1), which generate A_4 (Remark 4.8). Composing them preserves the evaluation, with every edge and vertex decoration transported. The inverse local rules (4.5)–(4.6) return each resulting decoration to the reference pattern. □

Now let us reformulate the TPE in the spherical case using these new notations.

Theorem 4.18 (Spherical TPE)**.** *Assume that the pivotal fusion category $\mathcal{C}$ of Theorem 4.12 is spherical. With the same objects, morphisms and dual bases for the evaluation pairing, the following identity holds:*

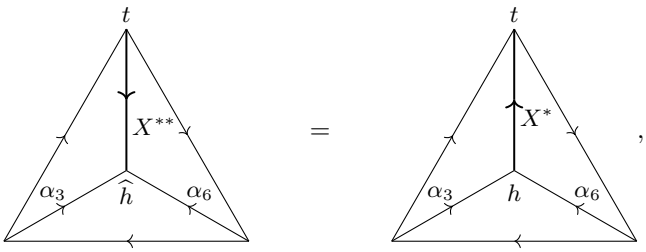


Proof. For the first left factor write $(u, v, w) = (\rho^{-2}(\alpha_2), \rho(\alpha_3), \rho^{-1}(\alpha_1))$. Rotate the reference tetrahedron $T(u, v, w, \beta_0)$, fixing its left vertex, then use (4.5)–(4.6) at the marked corners:



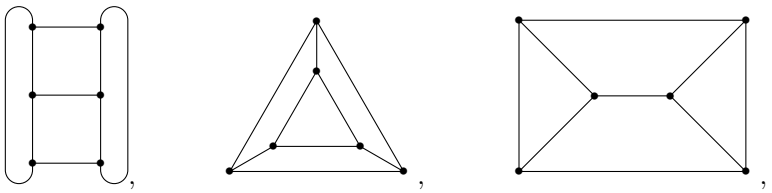
In the last move $v \mapsto \rho^{-1}(v)$, $u \mapsto \rho^2(u)$ and $w \mapsto \sigma^{-2}(\rho(w))$: the base edge reverses at the left corner and reverses back at the right.

For the first right factor put $t = \rho^2(\beta_1)$, $h = \rho(\beta_3')$ and $\widehat{h} = \rho(\sigma(\beta_3'))$. In its reference graph apply ρ^{-1}, ρ, ρ^3 at the left, right and central vertices, respectively. Only the middle edge changes in the following equality; the common outer edge labels are suppressed:



since $\widehat{h} = (\mathrm{id}_{X_8^*} \otimes a_{X^*} \otimes \mathrm{id}_{X_5^{**}}) \circ h$ and $\mathrm{ev}_{X^*,-} = \mathrm{ev}_{X^{**},+} \circ (a_{X^*} \otimes \mathrm{id}_{X^{**}})$. Apply the same corner rules to the remaining factors of Theorem 4.12, using the paired bases of Lemma 3.1. □

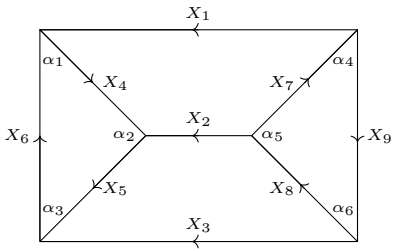
Remark 4.19. Here are three representations of the triangular prism graph,



the first is the one we used in Definition 4.9, the second is the usual one (already mentioned in §1), and the last is the one we will use in Definition 4.20.

Following Remark 4.10, we can realize the orientation-preserving symmetry group S_3 of the regular triangular prism, as for the regular tetrahedron in Remark 4.16. The monoidal triangular prism in Definition 4.9 can be represented pictorially (as for Definition 4.13) using Definition 4.20.

Definition 4.20. Following Definition 4.9,



$:= TP(\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6).$

Remark 4.21. One may similarly compare tetrahedral decompositions of other planar trivalent graphs, for example the cubical graph. As in Thurston's prism picture (Figure 1), each decomposition gives a contraction of tetrahedral coefficients. When the graph has a regular geometric realization, its orientation-preserving symmetries act by transporting the edge and vertex decorations. These symmetries and the choice of cuts may expose useful local equations for categorification problems. In a spherical monoidal category the resulting identities follow from associativity and duality; §5.2 identifies the triangular prism equation itself with the pentagon relation after a change of coordinates. A reconstruction from scalar solutions must in addition impose associator invertibility, unit normalization, and compatible duality and spherical data. The smallest non-planar trivalent graph gives the braided variant discussed in Remark 4.26.

Remark 4.22. By [29, Definition 2.1 and Theorem 2.2, p. 70], every spherical monoidal category is pivotally equivalent to a strictly pivotal spherical category. A pivotal equivalence transports both pivotal traces under its identification of endomorphisms of the unit, and therefore preserves sphericality [29, Lemma 6.1, p. 78]. In that realization $a_X = \mathrm{id}_X$ for every object X, the duality functor has identity monoidal constraints, and $(-)^{**} = \mathrm{Id}$ on both objects and morphisms. Hence $\sigma = \mathrm{id}$. In the fusion setting we take the equivalence to be $\Bbbk$-linear, so the categorical dimensions agree as scalars in $\Bbbk$. A tensor equivalence preserves fusion multiplicities, and a $\Bbbk$-linear pivotal equivalence preserves the numerical Frobenius–Schur indicators [29, Corollary 4.4, p. 74]. We may therefore use this realization for the following simplified TPE and the calculations that follow without changing the categorification problem.

The realization need not be skeletal: self-duality means $X \simeq X^*$, not equality with the chosen dual. For a simple self-dual X and an isomorphism $\kappa : X \to X^*$, (2.4) specializes to $\kappa^* = \nu_2(X)\kappa$. Transporting the duality along κ to make X its own chosen dual gives pivotal component $\kappa^{-1}\kappa^* = \nu_2(X)\mathrm{id}_X$, as in Proposition 2.7.

In the strictly pivotal realization of Remark 4.22, reverse the edges X_i for $i = 1, 2, 3, 7, 8, 9$ of Theorem 4.18 and use the dual objects as labels (Figure 2). The corner changes now involve only ρ and ρ^{-1}. With a general pivotal structure the same changes retain the pivotal morphisms in (4.6).

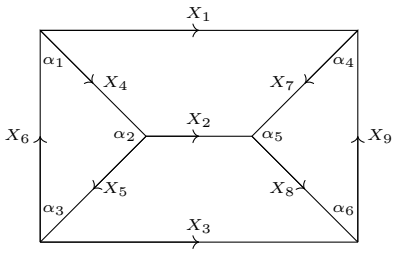
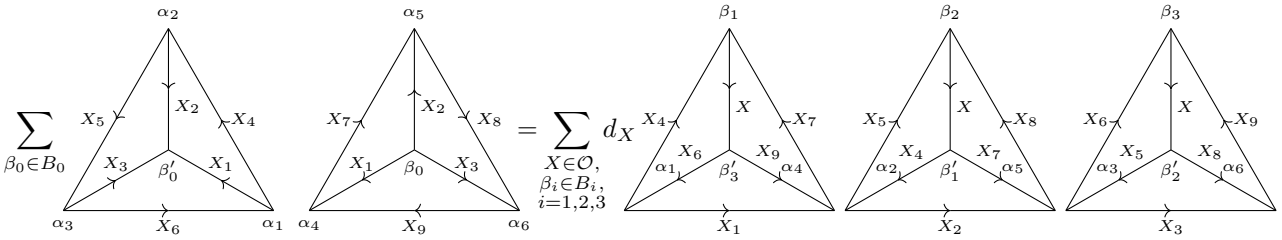


FIGURE 2. Triangular prism configuration

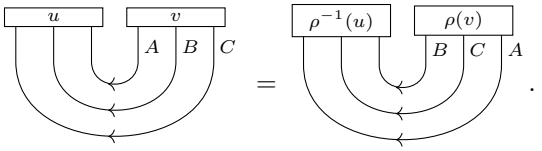
Theorem 4.23 (Simplified TPE)**.** *Let $\mathcal{C}$ be a spherical fusion category in the strictly pivotal realization of Remark 4.22; thus $a_X = \mathrm{id}_X$ for every object and $(-)^{**} = \mathrm{Id}$ on objects and morphisms. Let $X_1, \ldots, X_9$ be objects in $\mathcal{C}$. Consider morphisms $\alpha_1 \in \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X_4 \otimes X_1 \otimes X_6^*)$, $\alpha_2 \in \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X_5 \otimes X_2 \otimes X_4^*)$, $\alpha_3 \in \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X_6 \otimes X_3 \otimes X_5^*)$, $\alpha_4 \in \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X_9^* \otimes X_1^* \otimes X_7)$, $\alpha_5 \in \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X_7^* \otimes X_2^* \otimes X_8)$ and $\alpha_6 \in \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X_8^* \otimes X_3^* \otimes X_9)$. Then the following*

TPE holds:



where $d_X = \dim(X)$; $\mathcal{O}$ is the set of simple subobjects of all three objects $X_4 \otimes X_7$, $X_5 \otimes X_8$ and $X_6 \otimes X_9$ (up to isomorphism); B_i is a basis of $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, Z_i)$ with $Z_0 = X_3 \otimes X_2 \otimes X_1$, $Z_1 = X_4^* \otimes X \otimes X_7^*$, $Z_2 = X_5^* \otimes X \otimes X_8^*$ and $Z_3 = X_6^* \otimes X \otimes X_9^*$. Let B_i' be the basis of $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, Z_i^*)$ dual to B_i under the pairing of Lemma 3.1. For $\beta_i \in B_i$, let β_i' denote the corresponding element of B_i'. For $i = 1, 2, 3$, write $Z_i(X), B_i(X), B_i'(X)$ when making their dependence on the summation variable explicit; B_0 is fixed. The set $\mathcal{O}$ contains one representative of each common simple constituent.

Proof. Substitute $(X_1^*, X_2^*, X_3^*, X_4, X_5, X_6, X_7^*, X_8^*, X_9^*)$ in Theorem 4.18. Strict pivotality gives $\sigma = \rho^3 = \mathrm{id}$. To see how the dual bases change, let $v : \mathbf{1} \to A \otimes B \otimes C$ and $u : \mathbf{1} \to C^* \otimes B^* \otimes A^*$. The two drawings below evaluate the same spherical diagram, first with C outermost and then with A outermost:

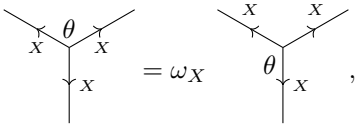


The left scalar is $\mathrm{ev}_{A\otimes B\otimes C} \circ (u \otimes v)$, the pairing of Lemma 3.1. Thus rotating a basis by ρ rotates its dual by ρ^{-1}. At the middle cut, (2.2) allows the two bases to be interchanged. In terms of the bases of the present theorem, those of Theorem 4.18 are therefore

j	0	1	2	3
basis element	β_0'	$\rho(\beta_3)$	$\rho(\beta_1)$	$\rho(\beta_2)$
dual element	β_0	$\rho^{-1}(\beta_3')$	$\rho^{-1}(\beta_1')$	$\rho^{-1}(\beta_2')$.

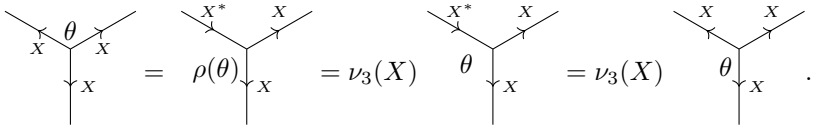
For instance, the second factor on the right of Theorem 4.18 has top label $\rho^2(\rho(\beta_1)) = \beta_1$ and central label $\rho(\rho^{-1}(\beta_3')) = \beta_3'$. Its upward X^* edge is the downward X edge of the first factor here, since $\mathrm{ev}_{X,-} = \mathrm{ev}_{X^*,+}$. The same substitutions give the two left factors in order and the right factors in order $(2, 3, 1)$, with the same categorical dimension d_X. □

Lemma 4.24 (Rotation eigenvalue)**.** *In the strictly pivotal realization of Remark 4.22, suppose $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X^{\otimes 3})$ is one-dimensional with generator θ. Then*



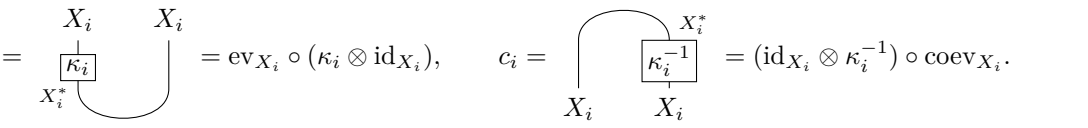
where $\omega_X := \nu_3(X)$, so that $\omega_X^3 = 1$.

Proof. Since $\sigma = \mathrm{id}$, Remark 4.3 gives $\rho(\theta) = \nu_3(X)\theta$. Apply (4.5), substitute this identity, and reverse the dual-labelled edge using $a_X = \mathrm{id}_X$ and Lemma 2.8:



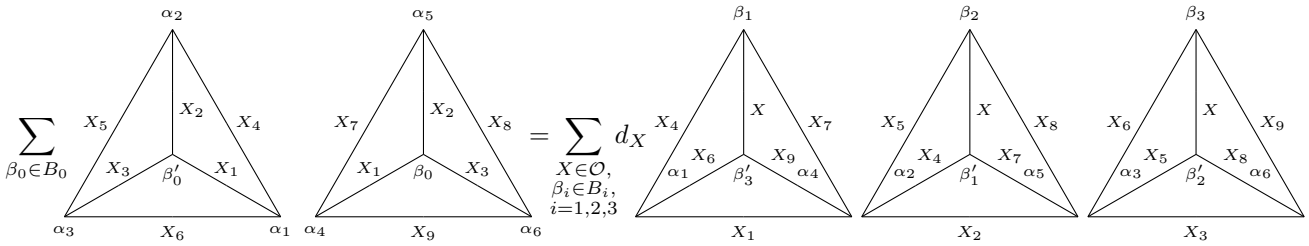
□

Remark 4.25 (More simplifications)**.** Work in the realization of Remark 4.22. Suppose that every external edge label and every internal label occurring in the sum is a self-dual simple object of second indicator $+1$. Then any chosen isomorphism $\kappa_i : X_i \to X_i^*$ satisfies $\kappa_i^* = \kappa_i$ by (2.4). Transport the cap and cup:

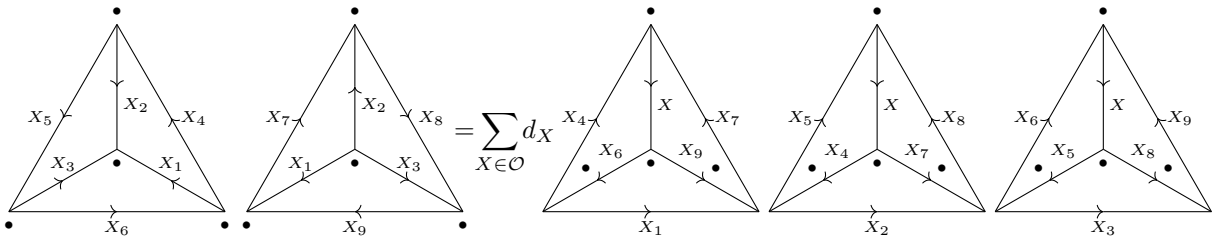


Omit arrows. These maps satisfy the zigzag identities since $\kappa_i^{-1} \circ \kappa_i = \mathrm{id}_{X_i}$, and $\kappa_i^* = \kappa_i$ identifies the two orientations of each cap or cup. Using the same isomorphisms at the vertices and in the evaluation pairings,

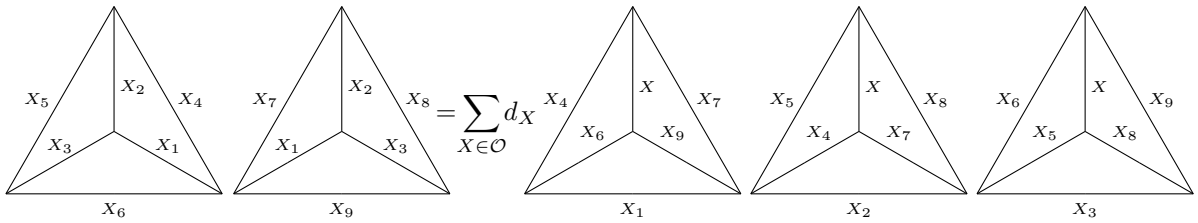
Theorem 4.23 reads



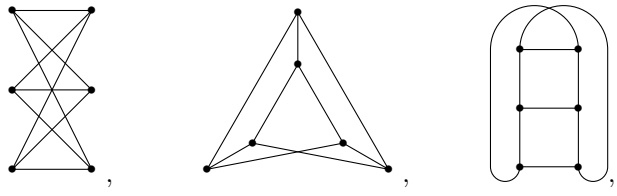
Omit basis indices. In the oriented calculus, suppose each nonzero vertex space is one-dimensional. Choose basis vectors at the external vertices. Each internal basis sum has at most one term. We set tetrahedral coefficients to zero when a vertex is inadmissible. The symbol • denotes the chosen vector or its dual for the evaluation pairing, in the indicated corner:



Omit arrows and corner marks. Use the symmetric κ_i above. Assume that the fusion rules are multiplicity-free and $\nu_3(X) = 1$ whenever $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X^{\otimes 3}) \neq 0$. Choose one basis vector in each cyclic orbit of vertex spaces and rotate it to the other corners. The only triples with nontrivial stabilizer are (X, X, X), where rotation acts trivially by Lemma 4.24. For two distinct orbits exchanged by reversing the order of the labels, use dual bases. For an orbit fixed by reversal, the pairing of a vector with its corresponding rotation is a nonzero scalar c. Rescaling by $c^{-1/2}$ makes the two vectors dual; the standing assumption that $\Bbbk$ is algebraically closed suffices here. The corner marks may thus be dropped:

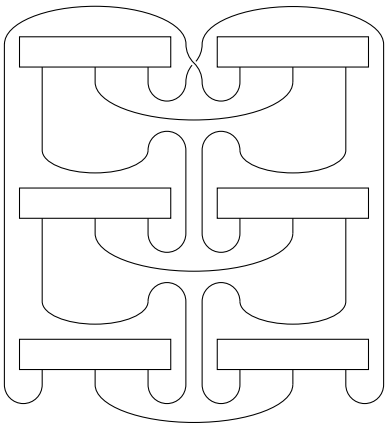


Remark 4.26. Let us conclude this section with some remarks on non-planar trivalent graphs and braiding. Besides the tetrahedral and triangular prism graphs, there exists a third connected simple trivalent graph with no more than six vertices, which is *non-planar*: the complete bipartite graph $K_{3,3}$. Below are three representations of this graph:



The first representation is the conventional one, the second is crafted to resemble the typical depiction of the triangular prism graph, and the third is used to derive the following monoidal category version, assuming the

presence of a *braiding*:



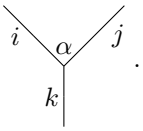
This illustration closely resembles the one in Definition 4.9, with the primary difference being the inclusion of braiding. It suggests a *braided version* of Theorem 4.12 (TPE). In general, we might derive such equations from any trivalent graph.

5. Pentagon and triangular prism equations

Notation 5.1. Let $(X_i)_{i\in I}$ represent the isomorphism classes of simple objects of a fusion category $\mathcal{C}$, with $X_1 = \mathbf{1}$. Write $N_{i,j}^k = \dim_{\Bbbk} \mathrm{Hom}_{\mathcal{C}}(X_i \otimes X_j, X_k)$ and abbreviate the edge label X_i to i. In the pivotal setting we write $d_i = \dim(X_i) = \dim_a(X_i)$ for the dimension defined by the pivotal structure a [11, Definition 4.7.11]. Selfduality means $X \simeq X^*$.

5.1. **Pentagon equations.** Choose a basis $B(i,j;k)$ of $\mathrm{Hom}_{\mathcal{C}}(X_i \otimes X_j, X_k)$. We draw $\alpha \in B(i,j;k)$ as a trivalent vertex:

(5.1)

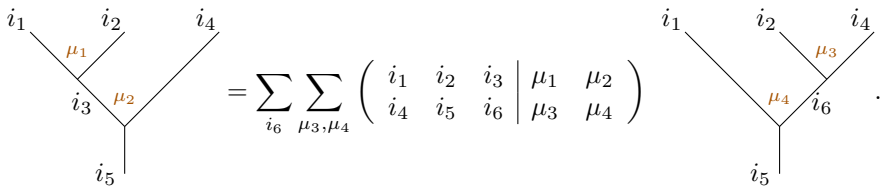


For fixed $\mu_1 \in B(i_1,i_2;i_3)$ and $\mu_2 \in B(i_3,i_4;i_5)$, the F-symbols are the coefficients of the change between the two bases of $\mathrm{Hom}_{\mathcal{C}}(X_{i_1} \otimes X_{i_2} \otimes X_{i_4}, X_{i_5})$ obtained by first decomposing the left or the right pair of factors [4, 8, 39]. With the associativity constraint understood, they are defined by

$$\mu_2 \circ (\mu_1 \otimes \mathrm{id}_{X_{i_4}}) = \sum_{i_6} \sum_{\mu_3,\mu_4} \left(\begin{array}{ccc|cc} i_1 & i_2 & i_3 & \mu_1 & \mu_2 \\ i_4 & i_5 & i_6 & \mu_3 & \mu_4 \end{array} \right) \mu_4 \circ (\mathrm{id}_{X_{i_1}} \otimes \mu_3),$$

where $i_6 \in I$, $\mu_3 \in B(i_2,i_4;i_6)$ and $\mu_4 \in B(i_1,i_6;i_5)$. In pictures,

(5.2)



Expanding the two routes of the pentagon in these bases gives the *pentagon equations* (PE)

(5.3)

$$\sum_{\mu_0} \left(\begin{array}{ccc|cc} i_2 & i_7 & i_8 & \mu_5 & \mu_6 \\ i_9 & i_3 & i_1 & \mu_4 & \mu_0 \end{array} \right) \left(\begin{array}{ccc|cc} i_5 & i_4 & i_2 & \mu_2 & \mu_0 \\ i_1 & i_3 & i_6 & \mu_1 & \mu_3 \end{array} \right)$$
$$= \sum_{i_0} \sum_{\mu_7,\mu_8,\mu_9} \left(\begin{array}{ccc|cc} i_5 & i_4 & i_2 & \mu_2 & \mu_5 \\ i_7 & i_8 & i_0 & \mu_7 & \mu_8 \end{array} \right) \left(\begin{array}{ccc|cc} i_5 & i_0 & i_8 & \mu_8 & \mu_6 \\ i_9 & i_3 & i_6 & \mu_9 & \mu_3 \end{array} \right) \left(\begin{array}{ccc|cc} i_4 & i_7 & i_0 & \mu_7 & \mu_9 \\ i_9 & i_6 & i_1 & \mu_4 & \mu_1 \end{array} \right).$$

Here $i_1, \ldots, i_9 \in I$, and the basis labels have the following types: $\mu_1 \in B(i_4,i_1;i_6)$, $\mu_2 \in B(i_5,i_4;i_2)$, $\mu_3 \in B(i_5,i_6;i_3)$, $\mu_4 \in B(i_7,i_9;i_1)$, $\mu_5 \in B(i_2,i_7;i_8)$, $\mu_6 \in B(i_8,i_9;i_3)$, $\mu_7 \in B(i_4,i_7;i_0)$, $\mu_8 \in B(i_5,i_0;i_8)$, $\mu_9 \in B(i_0,i_9;i_6)$ and $\mu_0 \in B(i_2,i_1;i_3)$, where $i_0 \in I$ is summed on the right. To read the identity, fix the bottom and upper-right trees of Figure 3. Expand the first in the basis containing the second by successive applications of (5.2). Along the two-move route the only free label is μ_0. Along the three-move route it is necessary to sum over i_0, μ_7, μ_8, μ_9. The pentagon axiom equates the two coefficients.

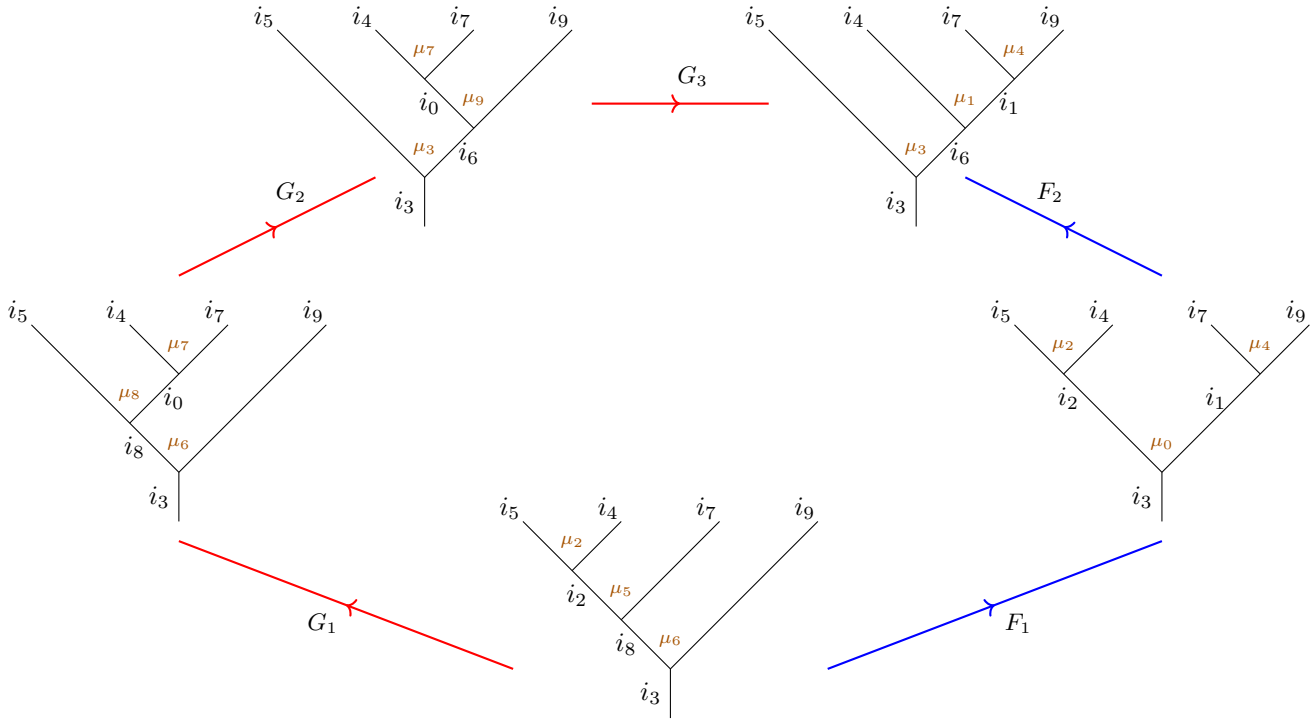


FIGURE 3. The two routes in the pentagon (5.3). Each arrow applies (5.2) once. Theorem 6.7 uses the blue route F_1, F_2, and Theorem 6.8 uses the red route G_1, G_2, G_3.

5.2. **TPE versus PE.** We first bend a fusion vertex into a three-legged invariant tensor. We then close the two sides of an F-move and follow the resulting tetrahedra through the pentagon.

Definition 5.2. Let $\mathcal{C}$ be a monoidal category with chosen left duals. Consider the functors $G, H : \mathcal{C}^{\mathrm{op}} \times \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to$ Set given by

$$G(X,Y,Z) = \mathrm{Hom}_{\mathcal{C}}(X \otimes Y, Z), \qquad H(X,Y,Z) = \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, Z \otimes Y^* \otimes X^*).$$

For $\mu : X \otimes Y \to Z$, the following formulas give the natural isomorphism $G \simeq H$ and its inverse, with $(X \otimes Y)^* = Y^* \otimes X^*$ as in §2:

(5.4)

$$\widetilde{\mu} = \quad = (\mu \otimes \mathrm{id}_{(X\otimes Y)^*}) \circ \mathrm{coev}_{X\otimes Y},$$

$$\mu = \quad = (\mathrm{id}_Z \otimes \mathrm{ev}_{X\otimes Y}) \circ (\widetilde{\mu} \otimes \mathrm{id}_{X\otimes Y}).$$

The zigzag identities make these assignments inverse to each other [11, Proposition 2.10.8].

For the remainder of §5.2, let $\mathcal{C}$ be a spherical fusion category in the strictly pivotal realization of Remark 4.22. Fix simple objects A, B, C. For each μ in a basis of $\mathrm{Hom}_{\mathcal{C}}(A \otimes B, C)$, let $q_\mu \in \mathrm{Hom}_{\mathcal{C}}(C, A \otimes B)$ be its dual for the trace pairing of §3.1, characterized by $\mathrm{tr}(\nu \circ q_\mu) = \delta_{\nu,\mu}$ for every ν in the basis. Bending q_μ gives

$$'\widetilde{\mu} = (q_\mu \otimes \mathrm{id}_{C^*}) \circ \mathrm{coev}_C \ \in \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, A \otimes B \otimes C^*). \tag{5.5}$$

Set $Z = C \otimes B^* \otimes A^*$. Closing the two tensors using the evaluation pairing b_Z of Lemma 3.1 gives

$$b_Z('\widetilde{\mu}, \widetilde{\nu}) = \mathrm{tr}(\nu \circ q_\mu) = b_{Z^*}(\widetilde{\nu}, '\widetilde{\mu}) = \delta_{\nu,\mu}.$$

Thus the primed basis in $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, Z^*)$ is dual to the bent basis in $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, Z)$ for b_Z. The categorical dual of the morphism $\widetilde{\mu}$ itself would have type $Z^* \to \mathbf{1}$.

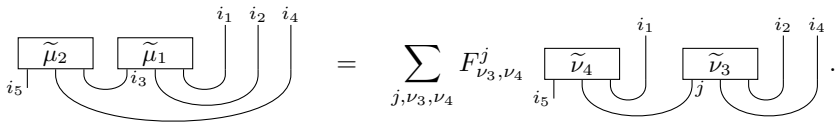
Theorem 5.3 (PE–TPE, change of basis)**.** *The triangular prism equation of Theorem 4.23 is the pentagon equation* (5.3) *under the following invertible change of coordinates. Write V_i for the simple representative indexed by i in Notation 5.1, to distinguish it from the numbered edges of the prism. For given pentagon labels, the prism edges are $X = V_{i_0}$ and*

$$(X_1, \dots, X_9) = (V_{i_3}^*, V_{i_1}, V_{i_2}, V_{i_6}, V_{i_4}, V_{i_5}^*, V_{i_9}^*, V_{i_7}, V_{i_8}).$$

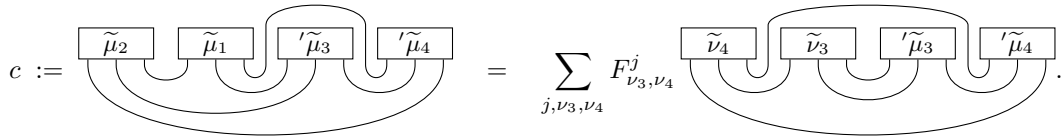
The table gives the vertex morphisms, all with source $\mathbf{1}$.

Vertex	Morphism	Target
α_1	$\rho({}'\widetilde{\mu}_3)$	$X_4 \otimes X_1 \otimes X_6^*$
α_2	${}'\widetilde{\mu}_1$	$X_5 \otimes X_2 \otimes X_4^*$
α_3	$\rho^{-1}(\widetilde{\mu}_2)$	$X_6 \otimes X_3 \otimes X_5^*$
α_4	$\rho^{-1}(\widetilde{\mu}_6)$	$X_9^* \otimes X_1^* \otimes X_7$
α_5	$\rho({}'\widetilde{\mu}_4)$	$X_7^* \otimes X_2^* \otimes X_8$
α_6	$\rho(\widetilde{\mu}_5)$	$X_8^* \otimes X_3^* \otimes X_9$
β_0	${}'\widetilde{\mu}_0$	$X_3 \otimes X_2 \otimes X_1$
β_1	$\rho^{-1}({}'\widetilde{\mu}_9)$	$X_4^* \otimes X \otimes X_7^*$
β_2	$\rho^{-1}(\widetilde{\mu}_7)$	$X_5^* \otimes X \otimes X_8^*$
β_3	${}'\widetilde{\mu}_8$	$X_6^* \otimes X \otimes X_9^*$.

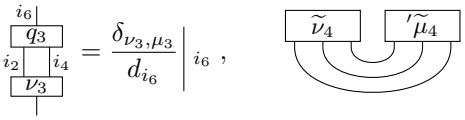
Proof. Extract the coefficient. Fix μ_1, μ_2 in (5.2) and write $F^j_{\nu_3,\nu_4}$ for its coefficient with intermediate label j and lower pair ν_3, ν_4. Bending the two input legs at each vertex by Definition 5.2 gives the same recoupling:



Choose the lower pair μ_3, μ_4 with intermediate label i_6 and let q_3, q_4 be its trace-duals. Attach ${}'\widetilde{\mu}_3$ and ${}'\widetilde{\mu}_4$ to the open legs and close the i_5 strand:



On the right, straighten the two legs joining the middle coupons. The resulting map $\nu_3 \circ q_3 : X_{i_6} \to X_j$ vanishes unless $j = i_6$. For $j = i_6$, Schur's lemma and the trace-dual normalization give

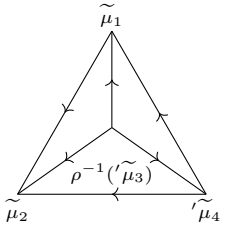


Thus the closure selects precisely the requested coefficient:

$$F^{i_6}_{\mu_3,\mu_4} = d_{i_6}\, c, \qquad c = \mathrm{tr}\big(\mu_2 \circ (\mu_1 \otimes \mathrm{id}_{X_{i_4}}) \circ (\mathrm{id}_{X_{i_1}} \otimes q_3) \circ q_4\big)\,. \tag{5.6}$$

Read the same closure as a tetrahedron. At ${}'\widetilde{\mu}_3$ start with its last leg $X^*_{i_6}$; this changes the marked corner by ρ^{-1}. The other three coupons of c already have the orders prescribed by Definition 4.4. In the reference pattern, the result is

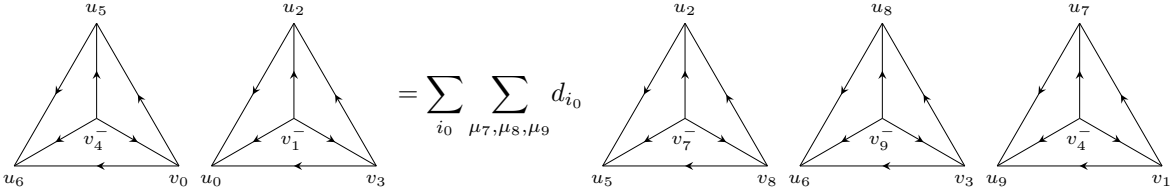
$$F^{i_6}_{\mu_3,\mu_4} = d_{i_6} \qquad = d_{i_6}\, T\big(\rho^{-1}({}'\widetilde{\mu}_3), \widetilde{\mu}_2, \widetilde{\mu}_1, {}'\widetilde{\mu}_4\big). \tag{5.7}$$



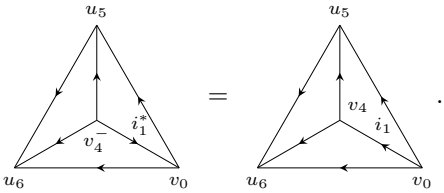
Substitute in the pentagon. By (5.7) the two factors on the left of (5.3) contribute $d_{i_1} d_{i_6}$, and the three factors on the right contribute $d_{i_0} d_{i_6} d_{i_1}$. Cancelling the common nonzero factor leaves d_{i_0}.

Write $u_j = \widetilde{\mu}_j$, $v_j = {}'\widetilde{\mu}_j$, and $w^{\pm} = \rho^{\pm 1}(w)$ for either kind of vertex label. The vertex types determine the edge labels. The final display marks the three edges labelled i_0. In the reference pattern of Definition 4.13, substitution in the pentagon gives

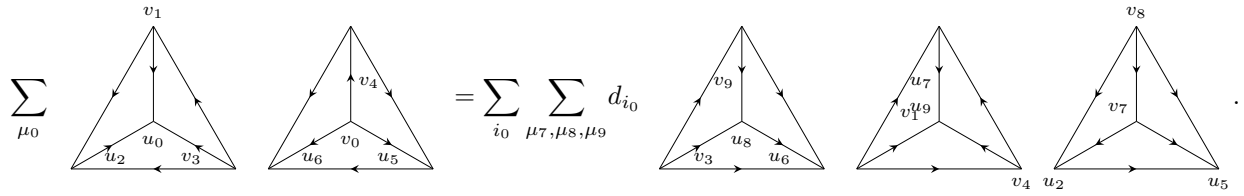
$$\sum_{\mu_0} \qquad = \sum_{i_0} \sum_{\mu_7,\mu_8,\mu_9} d_{i_0} \qquad . \tag{5.8}$$



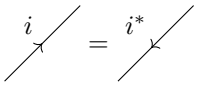
In each factor, move the central corner once by (4.5). The first left factor illustrates the move: its centre-to-right edge i_1^* becomes a right-to-centre edge i_1, and its central label changes from v_4^- to v_4. All other corners stay fixed:



Now rotate the five tetrahedra by the spherical symmetries of Proposition 4.17. Reorder the two left factors as $(2,1)$ and the three right factors as $(2,3,1)$. Rotating the arrows and marked corners with each tetrahedron gives

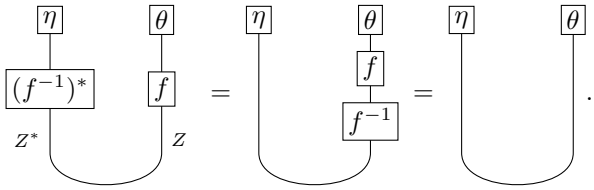


Match the corners and the paired bases. Apply (4.5)–(4.6) to place the labels at the corners of Theorem 4.23. With $a = \mathrm{id}$, the remaining edge reversals have the form



The fourth and fifth external vertices illustrate the corner convention: $\rho^{-1}(\widetilde{\mu})$ has ordered legs (A^*, C, B^*) and $\rho(\widetilde{\mu}')$ has ordered legs (B, C^*, A). Taking $\mu = \mu_6$ and $\mu = \mu_4$, respectively, gives the entries α_4 and α_5 of the table. Figure 4 gives the corresponding prism labels.

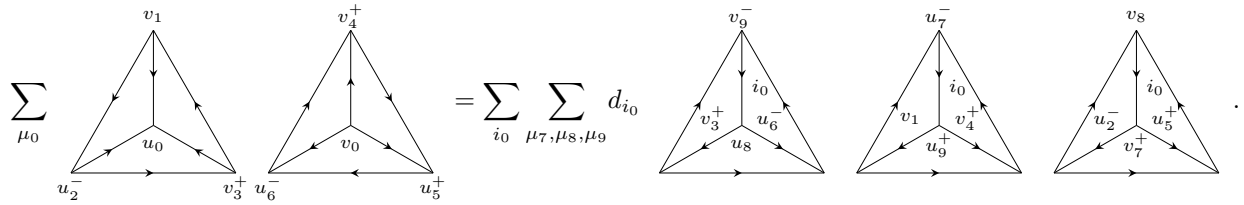
At an internal cut, the basis and its dual change together. Indeed, evaluating $(f^{-1})^* \circ \eta$ against $f \circ \theta$ gives the original pairing, for $f : Z \xrightarrow{\sim} Z$:



For the rotations in the table one uses the cyclic pairing identity from the proof of Theorem 4.23:

$$b_{B\otimes C\otimes A}\big(\rho^{-1}(u), \rho(v)\big) = b_{A\otimes B\otimes C}(u, v).$$

The resulting marked tetrahedra are



The centre of the first left factor is $u_0 = \beta_0'$, that of the second is $v_0 = \beta_0$. On the right, the three top labels are $(v_9^-, u_7^-, v_8) = (\beta_1, \beta_2, \beta_3)$, and the three centres are $(u_8, u_9^+, v_7^+) = (\beta_3', \beta_1', \beta_2')$. This is exactly the factor order and pairing in Theorem 4.23.

Multilinearity gives the identity for arbitrary vertex morphisms on simple edges. Resolving the identity on each external edge into projections and inclusions of simple summands then gives the identity for arbitrary edge objects. Conversely, the bends, rotations and paired basis changes are invertible, and the simple categorical dimensions are nonzero. Reading the calculation backwards recovers (5.3). □

Theorem 5.3 compares two expressions for associativity in a given spherical fusion category. Reconstruction from abstract scalars also requires invertible associativity matrices, unit normalization and compatible spherical duality data. The spherical localization arguments use these equations as necessary conditions. Their dimension factors d_i are categorical dimensions. They define a ring character [11, Proposition 4.7.12] and satisfy $d_i = d_{i^*}$ by sphericality.

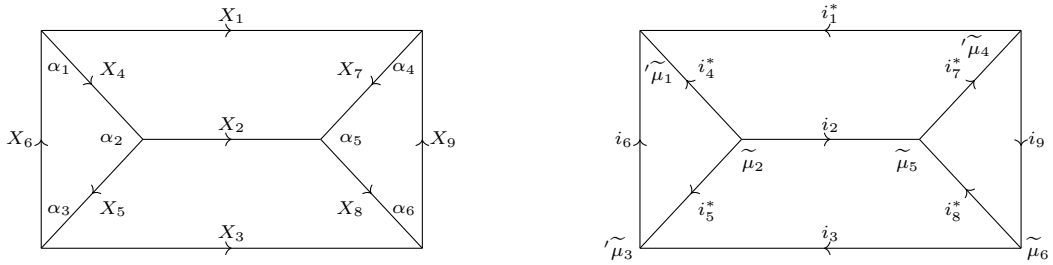


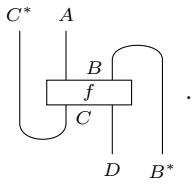
FIGURE 4. Left, the edge and vertex labels of the triangular prism equation (Figure 2). Right, the simple indices and bent basis morphisms of the pentagon equation (5.3). To pass between them, cyclically rotate each triangular end and apply the changes of marked corner in Theorem 5.3. For instance, α_1 at the upper left corresponds to $'\widetilde{\mu}_3$ at the lower left by $\alpha_1 = \rho('\widetilde{\mu}_3)$.

6. Localization

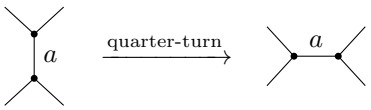
6.1. **Localization strategy.** Tetrahedral rotations identify coefficients, and rotations of the triangular prism identify equations among them. The groups are A_4 and $D_3 \cong S_3$, respectively. A rotation moves the orientations, dual labels and marked corners along with the graph. It therefore identifies decorated evaluations for which the ordinary pentagon would require explicit changes of basis.

The following pictures give a geometric interpretation of localization. Resolving $A \otimes B$ and $C \otimes D$ into simples gives a matrix-unit basis of $\mathrm{Hom}_{\mathcal{C}}(A \otimes B, C \otimes D)$. Its diagrams have the shape of an I. A quarter-turn turns the I into an H and gives the *string Fourier transform* [18]. For $f : A \otimes B \to C \otimes D$ it is the rigidity isomorphism

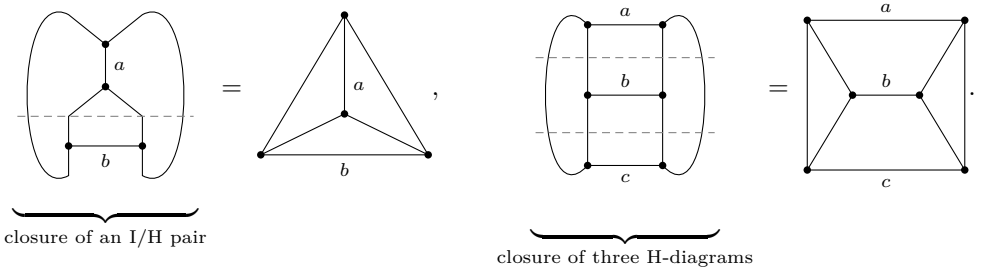
$$\mathrm{Hom}_{\mathcal{C}}(A \otimes B, C \otimes D) \simeq \mathrm{Hom}_{\mathcal{C}}(C^* \otimes A, D \otimes B^*), \qquad f \longmapsto \mathcal{F}(f) =$$



Here $\mathcal{F}(f) = (\mathrm{ev}_C \otimes \mathrm{id}_{D \otimes B^*}) \circ (\mathrm{id}_{C^*} \otimes f \otimes \mathrm{id}_{B^*}) \circ (\mathrm{id}_{C^* \otimes A} \otimes \mathrm{coev}_B)$. Suppressing the vertex bases and duality decorations of §§2–3, we obtain



Closing a pair of such diagrams gives a tetrahedron. Closing three gives a triangular prism. The dashed lines mark the cuts.



With the trace normalizations above, the tetrahedral coefficients are pairings of the quarter-turned basis vectors with the trace-normalized basis. Evaluating the closed prism before and after resolving its cuts gives the two sides of TPE. Repeating edge labels lets us group equations involving the same coefficients.

For a quantitative picture, suppose X is self-dual and $X \otimes X$ is multiplicity-free, with n simple summands. Then $\dim_{\Bbbk} \mathrm{End}_{\mathcal{C}}(X \otimes X) = n$. After the boundary identifications have been fixed, its quarter-turn map has n^2 matrix entries. Labelling the six edges of the two triangular ends by X leaves n^3 possible triples of simple labels on the three connecting edges. At the level of these triples, cyclic permutation has $(n^3 + 2n)/3$ orbits. This count describes the available configurations; it does not assert that all resulting equations are independent, or that their other tetrahedral contractions are already known. Unit labels and inadmissible vertices give explicit values for some contractions, while the others are retained as variables. The functions x and y in Theorem 6.1 record precisely these contractions.

The general localization procedure is iterative:

(1) Choose a measure of complexity for the unknown coefficients. For example, count the distinct simple labels, up to duality, occurring in a tetrahedron, and assign the lowest complexity to coefficients already known.

(2) Select a small collection of coefficients and equations involving them. Generate these collections incrementally; repeated-label prisms and their rotations supply systematic choices.
(3) Compute the consequences of the selected subsystem, using linear or binomial elimination when available and a Gröbner basis when needed.
(4) Substitute the resulting relations into further subsystems and add compatibility equations between overlapping choices. Retain every solution branch not ruled out by a necessary condition. An inconsistent subsystem already obstructs categorification; otherwise the process continues.

Useful subsystems can be much smaller than the full associator system. The multiplicity-free calculations in [34], for example, exploit binomial equations, equations linear in selected variables, and low-degree relations. The choice of subsystem remains important: a subsystem with many algebraic solution branches may be expensive to combine with the rest. The geometric construction supplies several families of related subsystems that can be compared, without assuming a uniform complexity bound for their solution. For the near-group fusion rules $G + |G|$, the two cuts give a sum of products of two coefficients on one side of the essential equation. He, Liu, Lu, Palcoux and Ren use this form to construct solutions for cyclic G under the spherical and indicator hypotheses recalled in the introduction [16, §2].

In the specializations used below, the first prism of Theorem 6.1 has k on all six edges of the triangular ends. Give the connecting edges labels $(a, b, 1)$ or (a, b, b). The second prism has (b, k, k) on each end and (k, k, a) on the connecting edges. On cutting these prisms, the same single tetrahedra and the same pairs recur. The braces in (6.3)–(6.4) show which evaluations to record as y and which contractions to record as x. Corollary 6.2 restricts a, b to S'_k while retaining the full internal sum over S_k. To compare two such choices for $\mathcal{F}_{210}$, Theorem 6.4 uses a prism whose nine edges carry just k and l.

The small-spectrum criteria in §6.3 have a different scope. They are derived directly from the pentagon equations and do not assume a pivotal or spherical structure. Thus the exclusion of $\mathcal{F}_{660}$ does not depend on the spherical reformulation.

6.2. **Categorification criteria from localization.** We use Notation 5.1. All categorical dimensions below belong to the algebraically closed ground field $\Bbbk$; the dimensions of simple objects are nonzero. The equations are necessary conditions for spherical categorification.

Theorem 6.1. *Let $\mathcal{C}$ be a spherical fusion category. Consider X_k, a self-dual simple object within $\mathcal{C}$, such that for every simple object X_a, the condition $N_{k,k}^a \le 1$ holds. Assume that if $N_{k,k}^a = 1$, then X_a is self-dual. Define S_k as the set of such indices a. Assume further that there exists b such that N_{b,b^*}^k is odd. There exist functions $x : S_k^3 \to \Bbbk$ and $y : S_k^2 \to \Bbbk$ satisfying, for every $a, b, c \in S_k$, the following equations:*

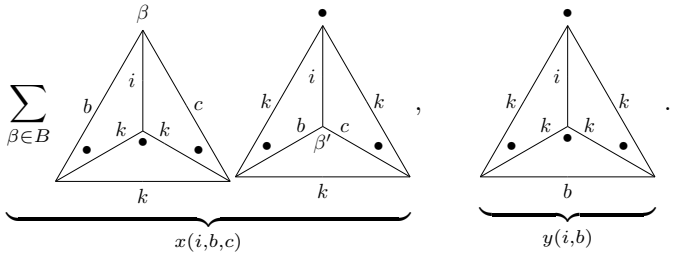
$$x(a,b,c) = \sum_{i \in S_k} d_i y(i,a) y(i,b) y(i,c), \tag{6.1}$$

$$y(a,b)y(a,c) = \sum_{i \in S_k} d_i y(i,a) x(i,b,c), \tag{6.2}$$

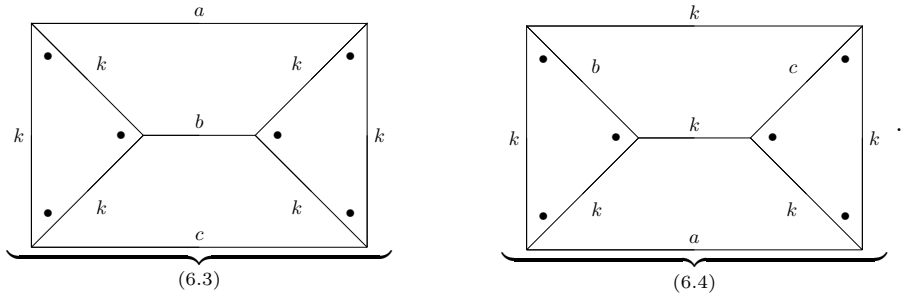
with $x(a,k,k) = y(a,k)^2$ whenever $k \in S_k$, $x(a,b,c) = 0$ if $N_{b,c}^a = 0$, $y(a,b) = y(b,a)$, $y(1,b) = d_k^{-1}$, $x(1,b,c) = \delta_{b,c}(d_b d_k)^{-1}$.

Proof. Work in the strictly pivotal realization of Remark 4.22. By Theorem 3.4, every X_i with $i \in S_k$ has indicator one (its multiplicity in $X_k \otimes X_k^*$ is one); the odd-multiplicity hypothesis gives the same conclusion for X_k. Fix symmetric self-duality maps κ_i as in Remark 4.25, with $\kappa_1 = \mathrm{id}_{\mathbf{1}}$, and use its unarrowed calculus throughout.

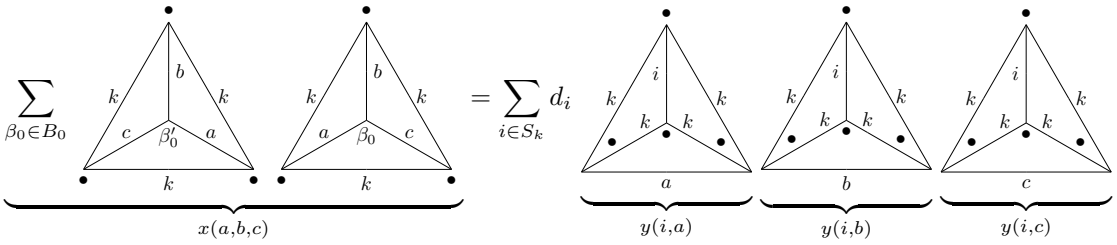
For each $i \in S_k$, choose a generator θ_i of the one-dimensional space $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X_k \otimes X_i \otimes X_k)$ with $\omega(\theta_i, \theta_i) = 1$, using the pairing of Lemma 3.5 under the self-duality identifications of Remark 4.25. The order (k, i, k) is unchanged by reversal, allowing this self-pairing. In the diagrams below, each $\bullet$ at a vertex of type (k, k, j) denotes the cyclic transport of θ_j determined by its marked corner. Opposite transports are dual, so the TPE introduces no additional normalization factors. For $i, b, c \in S_k$, let β run through a basis B of $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X_b \otimes X_i \otimes X_c)$, with dual vector β'. Define

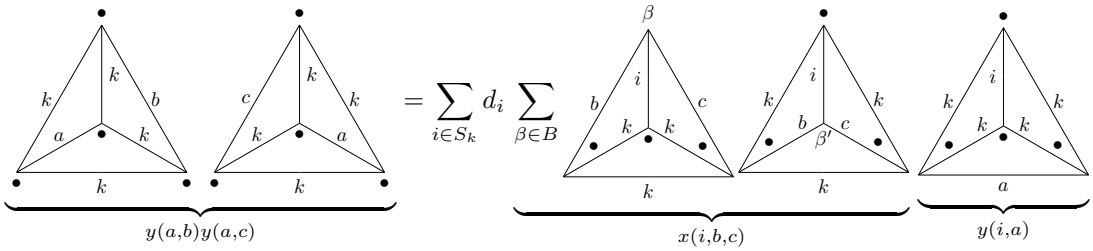


Apply Theorem 4.23 to the following prisms:

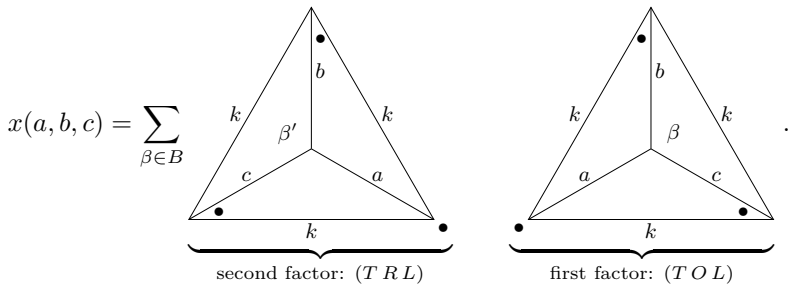


The three side cuts in the first prism have boundary object $X_k^{\otimes 2}$. In the second they have boundary objects $X_b \otimes X_c$ and two copies of $X_k^{\otimes 2}$. Thus we can sum over S_k in both equations: where the corresponding simple summand is absent from $X_b \otimes X_c$, the basis B is empty.

(6.3)


(6.4)


For the first left-hand side, write T, L, R, O for the top, left, right and interior vertices. Rotate the two factors defining $x(a,b,c)$ on the sphere and interchange them. Before restoring their marked corners this gives

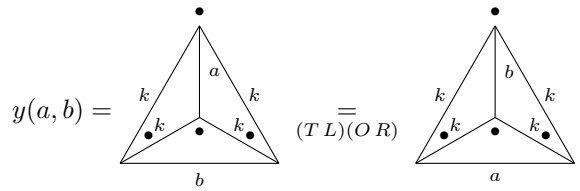


Moving the central marks to the lower face by (4.5) gives $\beta_0' = \rho(\beta')$ and $\beta_0 = \rho^{-1}(\beta)$, a pair of vectors dual under the evaluation pairing. By Lemma 4.24 the remaining corner factors multiply to

$$\underbrace{\omega_k^{\delta_{k,b}}\omega_k^{2\delta_{k,b}}}_{1}\ \underbrace{\omega_k^{2\delta_{k,c}}\omega_k^{\delta_{k,c}}}_{1} \qquad (\omega_k^3 = 1 \text{ if } k \in S_k;\ \omega_k := 1 \text{ otherwise}).$$

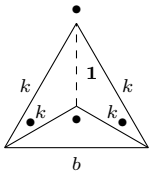
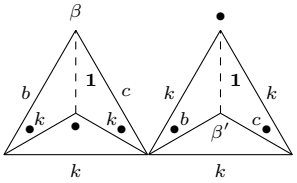
For the left side of (6.4), rotate $y(a,b)$ by $(T\,O\,L)$ and $y(a,c)$ by $(T\,O\,R)$. The two b-corner shifts are opposite, as are the two c-shifts; the a-shifts cancel between the factors. The underbraces now give (6.1) and (6.2).

The half-turn below exchanges TO and LR, with their marked corners:



If $k \in S_k$, then $N_{k,k}^a = 1$: in the definition of $x(a,k,k)$ the single pair is two copies of $y(a,k)$, with the same vector and its dual. Thus $x(a,k,k) = y(a,k)^2$.

Finally, an inadmissible vertex gives $x(a,b,c)=0$ when $N_{b,c}^{a}=0$. For the unit normalizations, the unit edges are dashed below; the paired unit vertices cancel by the dual-basis and zigzag identities:

$$y(1,b)=$$  $$=\frac{1}{d_k},\qquad x(1,b,c)=\sum_{\beta\in B}$$  $$=\frac{\delta_{b,c}}{d_b d_k}.$$

Here B is a basis of $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1},X_b\otimes X_c)$, and β' denotes the corresponding dual vector. This space vanishes unless $b=c$. For $b=c$, Schur's lemma converts the normalized dual-basis pairing into $d_b^{-1}\mathrm{id}_{X_b}$; the paired k-unit vertices contribute d_k^{-1}. □

Corollary 6.2. *Let $S_k'\subseteq S_k$. Restrict the functions of Theorem 6.1, and write $x(i,b)=x(i,b,b)$. There are functions $x(i,b):S_k\times S_k'\to\Bbbk$ and $y(i,b):S_k\times S_k'\to\Bbbk$ such that for all $a,b\in S_k'$*

$$\delta_{a,b}=d_b\sum_{i\in S_k}d_i y(i,a)y(i,b), \tag{6.5}$$

$$x(a,b)=\sum_{i\in S_k}d_i y(i,a)y(i,b)^2, \tag{6.6}$$

$$y(a,b)^2=\sum_{i\in S_k}d_i y(i,a)x(i,b), \tag{6.7}$$

with $x(a,k)=y(a,k)^2$ when $k\in S_k'$; $x(a,b)=0$ if $N_{b,b}^{a}=0$; $y(a,b)=y(b,a)$; $y(1,b)=d_k^{-1}$; $x(1,b)=(d_bd_k)^{-1}$.

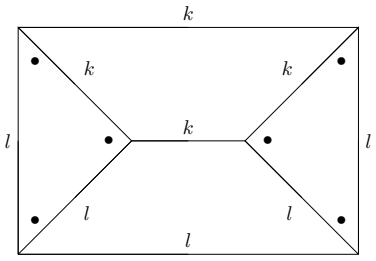
Proof. Since X_k is self-dual, $N_{k,k}^{1}=1$ and hence $1\in S_k$. The functions in Theorem 6.1 are defined on all of S_k, regardless of whether $1\in S_k'$. Equation (6.1) shows that $x(a,b,c)$ is symmetric in its three arguments. Set $c=1$ and use $y(i,1)=d_k^{-1}$ and $x(a,b,1)=\delta_{a,b}(d_bd_k)^{-1}$ to obtain (6.5). Setting $c=b$ in (6.1) and (6.2) gives (6.6) and (6.7). Restricting the second arguments to S_k' gives the stated subsystem. □

Remark 6.3. For an obstruction it suffices to retain any subset of these necessary equations. Fix an order on S_k'. One useful selection retains (6.5) for $a\neq 1$ and $a\geq b$, (6.6) for $a,b\neq 1$, and (6.7) for $b\notin\{1,k\}$. This is a selection of a necessary subsystem; no equivalence to the full system is needed for an obstruction. Symmetry avoids duplicating (6.5), and equations involving the unit often simplify after the prescribed values are substituted. In the application below, we state the retained equations explicitly. The general generator described in `GUIDE.md` constructs all instances and removes identically zero or scalar-proportional duplicates.

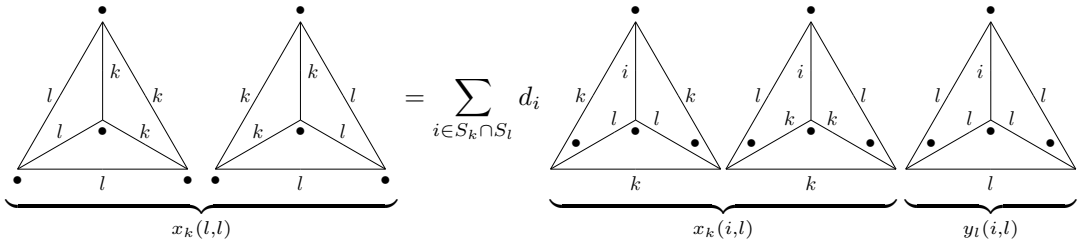
Theorem 6.4. *Suppose X_k and X_l both satisfy the hypotheses of Theorem 6.1, with $k\neq l$. Choose subsets $S_k'\subseteq S_k$ and $S_l'\subseteq S_l$ such that $l\in S_k'\cap S_l'$, and form the subsystems E_k,E_l of Corollary 6.2 using compatible vertex bases in the same category. Then their variables satisfy the additional equation*

$$x_k(l,l)=\sum_{i\in S_k\cap S_l}d_i y_l(i,l)x_k(i,l)$$

Proof. Use the same strictly pivotal realization, self-duality maps and marked vertex bases in the two localizations. Transport each basis together with its dual for the evaluation pairing, as in Theorem 6.1. The mixed prism is



The three side cuts carry $X_k^{\otimes 2},X_l^{\otimes 2},X_l^{\otimes 2}$. Hence $i\in S_k\cap S_l$ and $N_{k,k}^{i}=N_{l,l}^{i}=1$; also $N_{k,k}^{l}=N_{l,l}^{l}=1$ since $l\in S_k\cap S_l$. With $x_k(i,l)=x_k(i,l,l)$ the TPE reads

(6.8) 

The marks on the left are restored by spherical rotation as above; the two transports at the paired l, l, l vertices have inverse scalars. □

We apply these equations in §7.1.

Applying localization to another fusion ring. Choose k and S'_k as in Corollary 6.2. The `localize` command of `criteria.py` takes the ordered fusion coefficients, allows noncommutative fusion rings, and records the source of each polynomial. It accepts a rational categorical dimension character or treats its values as variables subject to

$$d_0 = 1, \qquad d_{i^*} = d_i, \qquad d_i d_j = \sum_a N_{ij}^a d_a, \qquad d_i r_i = 1$$

(the program numbers the unit 0). Inconsistency over $\overline{\mathbb{Q}}$ with symbolic dimensions excludes all spherical categorifications in characteristic zero. With prescribed dimensions it excludes only that character. For an integral fusion ring the Frobenius–Perron character suffices in characteristic zero by pseudo-unitarity [12, Proposition 8.24].

Use compatible vertex bases when coupling two systems by Theorem 6.4. Each distinguished label has its own variables, and an ordered pair (k, l) imposes the displayed compatibility equation. The file `GUIDE.md` gives the input format and worked examples.

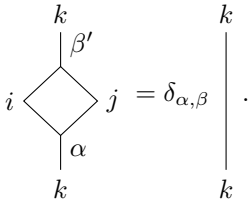
6.3. **Zero and one spectrum criteria.** Let $\Bbbk$ be any field and $\mathcal{C}$ a fusion category over $\Bbbk$, with $\mathrm{End}_{\mathcal{C}}(\mathbf{1}) = \Bbbk$, whose Grothendieck ring is the given fusion ring. Every simple X is split. Indeed, rigidity identifies $\mathrm{End}_{\mathcal{C}}(X)$ with $\mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X \otimes X^*)$, which is one-dimensional because the unit has multiplicity one. Thus the composition pairings and F-matrices of §5.1 are defined over $\Bbbk$ itself. The index i^* represents the common isomorphism class of the left and right duals of a simple. We use neither a pivotal structure nor categorical dimensions. Semisimplicity is essential to the resolution into simples, so the criteria do not address categorification by a nonsemisimple finite tensor category.

A summand on the right-hand side of (5.3) can occur only for

$$i_0 \in I_s = \{i \in I : N_{i_4,i_7}^i N_{i_5^*,i_8}^{i} N_{i_6,i_9^*}^{i} > 0\}.$$

We call this set the *spectrum* of the equation. An empty spectrum makes the three-factor side vanish. The first criterion forces the two-factor side to be nonzero. The second forces a unique nonzero term on the three-factor side and an empty sum on the two-factor side. All numerical conditions are tested in the nonnegative integers, in every characteristic.

Notation 6.5. For the basis $B(i, j; k)$ fixed in §5.1, choose the dual basis in $\mathrm{Hom}_{\mathcal{C}}(X_k, X_i \otimes X_j)$ for the composition pairing: $\alpha \circ \beta' = \delta_{\alpha,\beta} \mathrm{id}_{X_k}$. Equivalently,



The pairing is perfect: the projections onto the copies of X_k in $X_i \otimes X_j$ and their inclusions pair to the identity matrix. We write $\cup_i = \mathrm{ev}_{X_i}$ and $\cap_i = \mathrm{coev}_{X_i}$; the standard rigid adjunctions are used with their stated sources and targets.

Lemma 6.6 (One-dimensional trick)**.** *Choose basis morphisms $\mu_1 \in B(i_1, i_2; i_3)$, $\mu_2 \in B(i_3, i_4; i_5)$, $\mu_3 \in B(i_2, i_4; i_6)$ and $\mu_4 \in B(i_1, i_6; i_5)$. If*

$$\sum_{k \in I} N_{i_1,i_6}^k N_{i_3,i_4}^k = 1 \text{ or } \sum_{k \in I} N_{i_1,i_2}^k N_{i_5,i_4^*}^k = 1 \text{ or } \sum_{k \in I} N_{i_3,i_2^*}^k N_{i_5,i_6^*}^k = 1,$$

then

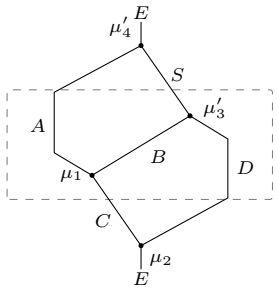
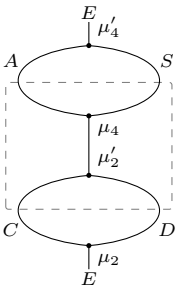
$$\left(\begin{array}{ccc|cc} i_1 & i_2 & i_3 & \mu_1 & \mu_2 \\ i_4 & i_5 & i_6 & \mu_3 & \mu_4 \end{array} \right) \neq 0.$$

Proof. Write $(A, B, C, D, E, S) = (X_{i_1}, \ldots, X_{i_6})$ and denote the displayed coefficient by F. By semisimplicity and rigidity, the three sums are, in order, the dimensions of

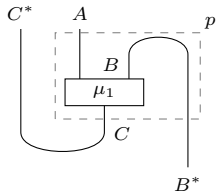
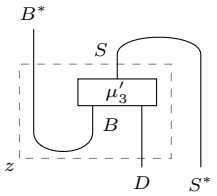
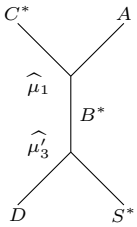
$$H_1 = \mathrm{Hom}_{\mathcal{C}}(A \otimes S, C \otimes D), \quad H_2 = \mathrm{Hom}_{\mathcal{C}}(A \otimes B \otimes D, E), \quad H_3 = \mathrm{Hom}_{\mathcal{C}}(E, C \otimes B^* \otimes S).$$

Primes denote the composition-duals of Notation 6.5.

First cut. Precompose the recoupling (5.2) with the tree splitting E by μ'_4 and then S by μ'_3. This gives the first equality. If $\dim H_1 = 1$, the boxed maps are proportional:

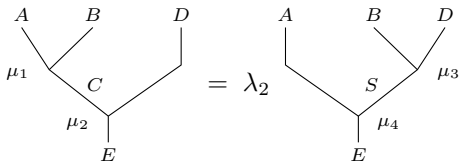
$F\,\mathrm{id}_E =$  $= \lambda_1$  $= \lambda_1\,\mathrm{id}_E.$

Both boxes are nonzero. The right one factors through the simple E by a split epimorphism followed by a split monomorphism. For the left one, bend its C- and S-legs by rigidity [11, Proposition 2.10.8]. Its mate factors through the simple B^* in the same way:

$\widehat{\mu}_1 =$  , $\widehat{\mu'_3} =$  ,  $\neq 0.$

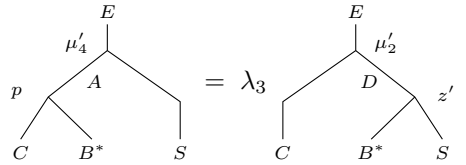
The dashed boxes here also define the one-bend mates $p : A \to C \otimes B^*$ and $z : B^* \otimes S \to D$. In particular $\lambda_1 \neq 0$.

Second cut. If $\dim H_2 = 1$, the two fusion trees are nonzero split epimorphisms, so

 $= \lambda_2$ $(\lambda_2 \neq 0).$

Precompose with $(\mathrm{id}_A \otimes \mu'_3) \circ \mu'_4$. The left tree gives $F\,\mathrm{id}_E$ and the right tree straightens to id_E, so $F = \lambda_2$.

Third cut. Choose a section z' of the nonzero map z onto D. If $\dim H_3 = 1$, the two splitting trees are nonzero split monomorphisms, so

 $= \lambda_3$ $(\lambda_3 \neq 0).$

Postcompose with $\mu_2 \circ (\mathrm{id}_C \otimes z)$. The zigzag between p and z straightens to the first network; on the right $z \circ z' = \mathrm{id}_D$ leaves $\mu_2 \circ \mu'_2 = \mathrm{id}_E$. Thus $F = \lambda_3$. □

Theorem 6.7 (Zero spectrum criterion)**.** *For a fusion ring R, if there exist indices $i_j \in I$ for $1 \le j \le 9$, such that the fusion coefficients $N_{i_4,i_1}^{i_6}$, $N_{i_5,i_4}^{i_2}$, $N_{i_5,i_6}^{i_3}$, $N_{i_7,i_9}^{i_1}$, $N_{i_2,i_7}^{i_8}$, and $N_{i_8,i_9}^{i_3}$ are non-zero, and if the following conditions hold:*

$$\sum_k N_{i_4,i_7}^k N_{i_5^*,i_8}^k N_{i_6,i_9^*}^k = 0, \tag{6.9}$$

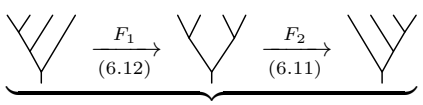
$$N_{i_2,i_1}^{i_3} = 1, \tag{6.10}$$

$$\sum_k N_{i_5,i_6}^k N_{i_2,i_1}^k = 1 \text{ or } \sum_{k\in I} N_{i_5,i_4}^k N_{i_3,i_1^*}^k = 1 \text{ or } \sum_k N_{i_2,i_4^*}^k N_{i_3,i_6^*}^k = 1, \tag{6.11}$$

$$\sum_k N_{i_2,i_1}^k N_{i_8,i_9}^k = 1 \text{ or } \sum_{k\in I} N_{i_2,i_7}^k N_{i_3,i_9^*}^k = 1 \text{ or } \sum_k N_{i_8,i_7^*}^k N_{i_3,i_1^*}^k = 1, \tag{6.12}$$

then R is not the Grothendieck ring of a fusion category over any field.

Proof. Choose the six external basis vectors. In the trees of Figure 3 the input labels are i_5, i_4, i_7, i_9, in that order, and the output is i_3. Read each row as the product of its change-of-basis coefficients. By (6.10) the two-move route has just one choice of μ_0. Lemma 6.6 makes each coefficient nonzero by the condition below its arrow. The three-move route has empty spectrum by (6.9). Thus the pentagon would give

 $= \underbrace{0}_{I_s=\varnothing}.$

□

Theorem 6.8 (One spectrum criterion)**.** *For a fusion ring R, if there are indices $i_j \in I$, $0 \le j \le 9$, such that the fusion coefficients $N_{i_4,i_1}^{i_6}$, $N_{i_5,i_4}^{i_2}$, $N_{i_5,i_6}^{i_3}$, $N_{i_7,i_9}^{i_1}$, $N_{i_2,i_7}^{i_8}$, $N_{i_8,i_9}^{i_3}$ are non-zero, and*

$$\sum_k N_{i_4,i_7}^k N_{i_5^*,i_8}^k N_{i_6,i_9^*}^k = 1, \tag{6.13}$$

$$N_{i_4,i_7}^{i_0} = N_{i_5^*,i_8}^{i_0} = N_{i_6,i_9^*}^{i_0} = 1, \tag{6.14}$$

$$N_{i_2,i_1}^{i_3} = 0, \tag{6.15}$$

$$\sum_k N_{i_5,i_0}^k N_{i_2,i_7}^k = 1 \text{ or } \sum_{k\in I} N_{i_5,i_4}^k N_{i_8,i_7^*}^k = 1 \text{ or } \sum_k N_{i_2,i_4^*}^k N_{i_8,i_0^*}^k = 1, \tag{6.16}$$

$$\sum_k N_{i_5,i_6}^k N_{i_8,i_9}^k = 1 \text{ or } \sum_{k\in I} N_{i_5,i_0}^k N_{i_3,i_9^*}^k = 1 \text{ or } \sum_k N_{i_8,i_0^*}^k N_{i_3,i_6^*}^k = 1, \tag{6.17}$$

$$\sum_k N_{i_4,i_1}^k N_{i_0,i_9}^k = 1 \text{ or } \sum_{k\in I} N_{i_4,i_7}^k N_{i_6,i_9^*}^k = 1 \text{ or } \sum_k N_{i_0,i_7^*}^k N_{i_6,i_1^*}^k = 1, \tag{6.18}$$

then R is not the Grothendieck ring of a fusion category over any field.

Proof. Fix the external basis vectors as above. By (6.15) the two-move route has no μ_0. Equations (6.13)–(6.14) leave just one term on the three-move route, with all three coefficients nonzero by Lemma 6.6. The pentagon would give

$$\underbrace{0}_{B(i_2,i_1;i_3)=\varnothing} = \underbrace{\xrightarrow[(6.16)]{G_1} \xrightarrow[(6.17)]{G_2} \xrightarrow[(6.18)]{G_3}}_{G_1G_2G_3\neq 0}.$$

□

Applying the spectrum criteria. The `spectrum` command of `criteria.py` tests the ordered fusion coefficients, including those of noncommutative fusion rings. It returns an ordered index tuple satisfying the chosen criterion, together with the integers in each hypothesis, or counts all such tuples. The search program calls each such tuple a *witness*. If no tuple satisfies either criterion, these tests give no obstruction. We apply the zero spectrum criterion to $\mathcal{F}_{660}$ in §7.2.

Proposition 6.9. *Neither spectrum criterion implies the other.*

Proof. Number the basis elements in each of the following examples from 1 to 6, with the unit first. Consider the following two rank-6 fusion data drawn from the dataset presented in [36]:

$$\begin{matrix}1&0&0&0&0&0\\0&1&0&0&0&0\\0&0&1&0&0&0\\0&0&0&1&0&0\\0&0&0&0&1&0\\0&0&0&0&0&1\end{matrix}\quad
\begin{matrix}0&1&0&0&0&0\\1&0&0&0&0&0\\0&0&1&0&0&0\\0&0&0&1&0&0\\0&0&0&0&1&0\\0&0&0&0&0&1\end{matrix}\quad
\begin{matrix}0&0&1&0&0&0\\0&0&1&0&0&0\\1&1&2&1&0&1\\0&0&0&1&0&0\\0&0&0&2&1&0\\0&0&1&1&0&0\end{matrix}\quad
\begin{matrix}0&0&0&1&0&0\\0&0&0&1&0&0\\0&0&1&2&2&1\\0&0&0&2&1&0\\0&0&2&1&1&1\\0&0&1&1&1&0\end{matrix}\quad
\begin{matrix}0&0&0&0&1&0\\0&0&0&0&1&0\\0&0&0&2&1&0\\1&1&1&1&0&0\\0&0&0&1&0&1\\0&0&1&1&1&0\end{matrix}\quad
\begin{matrix}0&0&0&0&0&1\\0&0&0&0&0&1\\0&0&1&1&0&0\\0&0&0&1&0&1\\1&1&0&0&1&0\\0&0&1&1&1&0\end{matrix}$$

$$\begin{matrix}1&0&0&0&0&0\\0&1&0&0&0&0\\0&0&1&0&0&0\\0&0&0&1&0&0\\0&0&0&0&1&0\\0&0&0&0&0&1\end{matrix}\quad
\begin{matrix}0&1&0&0&0&0\\1&0&0&0&0&0\\0&0&1&0&0&0\\0&0&0&1&0&0\\0&0&0&0&0&1\\0&0&0&0&1&0\end{matrix}\quad
\begin{matrix}0&0&1&0&0&0\\0&0&1&0&0&0\\1&1&2&0&0&0\\0&0&0&1&2&0\\0&0&0&2&1&1\\0&0&0&0&1&1\end{matrix}\quad
\begin{matrix}0&0&0&1&0&0\\0&0&0&1&0&0\\0&0&0&0&1&1\\1&1&0&2&0&0\\0&0&1&0&1&1\\0&0&1&1&1&2\end{matrix}\quad
\begin{matrix}0&0&0&0&1&0\\0&0&0&0&0&1\\0&0&0&1&1&1\\0&0&1&0&1&1\\1&0&1&1&0&2\\0&1&1&1&2&0\end{matrix}\quad
\begin{matrix}0&0&0&0&0&1\\0&0&0&0&1&0\\0&0&0&1&1&1\\0&0&1&0&1&1\\0&1&1&1&2&0\\1&0&1&1&0&2\end{matrix}$$

Counting the ordered tuples satisfying the zero- and one-spectrum criteria, in that order, gives $(12, 0)$ for the first fusion ring and $(0, 96)$ for the second. For the first fusion ring, $(i_1, \dots, i_9) = (4, 5, 6, 3, 5, 6, 4, 5, 6)$ satisfies Theorem 6.7; for the second, $(i_0, \dots, i_9) = (1, 5, 5, 5, 3, 4, 5, 3, 4, 5)$ satisfies Theorem 6.8. □

Proposition 6.10. *The smallest rank at which either spectrum criterion gives an obstruction is four. Both criteria attain this bound on the same fusion ring.*

Proof. No ordered tuple satisfies either criterion in ranks at most three, as shown in the supplementary proof and its finite check using the rank-three tables of [30, Proposition 3.1].

In rank four, number the basis from 1 to 4, with the unit first, and take

$$\begin{pmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{pmatrix},\quad
\begin{pmatrix}0&1&0&0\\1&1&0&1\\0&0&1&1\\0&1&1&2\end{pmatrix},\quad
\begin{pmatrix}0&0&1&0\\0&0&1&1\\1&1&1&0\\0&1&0&2\end{pmatrix},\quad
\begin{pmatrix}0&0&0&1\\0&1&1&2\\0&1&0&2\\1&2&2&3\end{pmatrix}.$$

Theorems 6.7 and 6.8 apply to the respective tuples

$$(i_1, \dots, i_9) = (3, 3, 2, 2, 3, 4, 3, 3, 3), \qquad (i_0, \dots, i_9) = (3, 4, 3, 3, 2, 3, 3, 3, 3, 2).$$

□

7. Applications

We apply the localization equations to $\mathcal{F}_{210}$ and the zero-spectrum criterion to $\mathcal{F}_{660}$. Together with the census recalled in the introduction, these exclusions prove Theorem 1.2. The verification guide lists the input data, scripts and certificates for both applications.

7.1. **Fusion ring $\mathcal{F}_{210}$.** The simple integral fusion ring $\mathcal{F}_{210}$ has rank 7, FPdim 210 and type $[[1,1],[5,3],[6,1],[7,2]]$, and the following fusion matrices:

$$\begin{matrix}1&0&0&0&0&0&0\\0&1&0&0&0&0&0\\0&0&1&0&0&0&0\\0&0&0&1&0&0&0\\0&0&0&0&1&0&0\\0&0&0&0&0&1&0\\0&0&0&0&0&0&1\end{matrix},\quad \begin{matrix}0&1&0&0&0&0&0\\1&1&0&1&0&1&1\\0&0&1&0&1&1&1\\0&1&0&0&1&1&1\\0&0&1&1&1&1&1\\0&1&1&1&1&1&1\\0&1&1&1&1&1&1\end{matrix},\quad \begin{matrix}0&0&1&0&0&0&0\\0&0&1&0&1&1&1\\1&1&1&0&0&1&1\\0&0&0&1&1&1&1\\0&1&0&1&1&1&1\\0&1&1&1&1&1&1\\0&1&1&1&1&1&1\end{matrix},\quad \begin{matrix}0&0&0&1&0&0&0\\0&1&0&0&1&1&1\\0&0&0&1&1&1&1\\1&0&1&1&0&1&1\\0&1&1&0&1&1&1\\0&1&1&1&1&1&1\\0&1&1&1&1&1&1\end{matrix},\quad \begin{matrix}0&0&0&0&1&0&0\\0&0&1&1&1&1&1\\0&0&1&1&1&1&1\\0&0&1&1&0&1&1\\1&1&1&1&1&1&1\\0&1&1&1&1&2&1\\0&1&1&1&1&2&2\end{matrix},\quad \begin{matrix}0&0&0&0&0&1&0\\0&1&1&1&1&1&1\\0&1&1&1&1&1&1\\0&1&1&1&1&2&1\\1&1&1&2&1&2&1\\0&1&1&1&2&1&2\\0&1&1&1&2&2&2\end{matrix},\quad \begin{matrix}0&0&0&0&0&0&1\\0&1&1&1&1&1&1\\0&1&1&1&1&1&1\\0&1&1&1&1&2&1\\0&1&1&1&2&1&2\\1&1&1&2&1&2&2\\1&1&1&2&2&1&1\end{matrix}$$

Remark 7.1. The interpolation construction helps explain why arithmetic categorification criteria do not exclude $\mathcal{F}_{210}$. The character table of $\mathrm{Rep}(\mathrm{PSL}(2,q))$ can be described uniformly, with separate formulas for q even and $q \equiv \pm 1 \pmod 4$ (see, e.g., [15, §5.2] or [9, §12.5]). This table can be interpolated to any integer $q > 1$, and applying the Schur orthogonality relations then yields an infinite family of simple integral fusion rings (simple for $q \geq 4$). When q is a prime power, these coincide with the Grothendieck ring of $\mathrm{Rep}(\mathrm{PSL}(2,q))$, but for other integers they give new fusion rings with the same favorable arithmetic properties (see [22]). The case $q = 6$ corresponds precisely to $\mathcal{F}_{210}$. It is expected that similar interpolated simple integral fusion rings exist for every family of finite simple groups of Lie type.

Let us label (and order) the simple objects by $1, 5_1, 5_2, 5_3, 6_1, 7_1, 7_2$, with FPdim $1, 5, 5, 5, 6, 7, 7$ (respectively). Observe that all the fusion matrices are self-adjoint, so the simple objects are self-dual.

Theorem 7.2. *The fusion ring $\mathcal{F}_{210}$ cannot be categorified in characteristic zero.*

Proof. A categorification in characteristic zero is defined over a finitely generated field by finitely many structure constants. Embedding that field into $\mathbb{C}$ preserves the fusion rules, so it suffices to work over $\mathbb{C}$. Suppose that there is a fusion category $\mathcal{C}$ over $\mathbb{C}$ whose Grothendieck ring is $\mathcal{F}_{210}$. Since $\mathcal{C}$ is integral, it is pseudo-unitary and then spherical by [11, Propositions 9.6.5 and 9.5.1]. We can apply Corollary 6.2 with $k = 5_1$, $S_k = \{1, 5_1, 5_3, 7_1, 7_2\}$, and $S'_k = \{1, 5_1, 5_3\}$. We get the following subsystem E_k of 10 variables and 12 equations:

$$\begin{aligned}
5u_0 + 7u_1 + 7u_2 - 4/25 &= 0, & 5v_1 + 5v_2 + 7v_4 + 7v_6 + 1/5 &= 0,\\
5v_0 + 5v_1 + 7v_3 + 7v_5 + 1/5 &= 0, & 25v_0v_1 + 25v_1v_2 + 35v_3v_4 + 35v_5v_6 + 1/5 &= 0,\\
25v_0^2 + 25v_1^2 + 35v_3^2 + 35v_5^2 - 4/5 &= 0, & 5v_0^2v_1 + 5v_1^2v_2 + 7v_3^2v_4 + 7v_5^2v_6 - v_1^2 + 1/125 &= 0,\\
5v_0^3 + 5v_1^3 + 7v_3^3 + 7v_5^3 - v_0^2 + 1/125 &= 0, & 25v_1^2 + 25v_2^2 + 35v_4^2 + 35v_6^2 - 4/5 &= 0,\\
5v_0v_1^2 + 5v_1v_2^2 + 7v_3v_4^2 + 7v_5v_6^2 + 1/125 &= 0, & 5v_1^3 + 5v_2^3 + 7v_4^3 + 7v_6^3 - u_0 + 1/125 &= 0,\\
5u_0v_1 - v_1^2 + 7u_1v_3 + 7u_2v_5 + 1/125 &= 0, & 5u_0v_2 - v_2^2 + 7u_1v_4 + 7u_2v_6 + 1/125 &= 0,
\end{aligned}$$

where $u_0 = x_k(5_3,5_3), u_1 = x_k(7_1,5_3), u_2 = x_k(7_2,5_3), v_0 = y_k(5_1,5_1), v_1 = y_k(5_3,5_1), v_2 = y_k(5_3,5_3), v_3 = y_k(7_1,5_1), v_4 = y_k(7_1,5_3), v_5 = y_k(7_2,5_1), v_6 = y_k(7_2,5_3)$.

For each local system, the selected prisms give 55 marked tetrahedral coefficients in 19 distinct nonzero equations when arrows are retained and unit edges are left unevaluated (count on GitHub). Rotating the tetrahedra, evaluating unit edges and taking the paired contractions that define x reduce the calculation to the 12 equations in 10 variables displayed above.

In the displayed system three linear equations eliminate u_0, v_0, v_2. The remaining seven variables admit an exact parametrization by the roots of a squarefree polynomial of degree 14. Thus there are exactly 14 complex solutions, as certified by `certificates/f210-solutions.json` and the spanning bound below.

Next we can apply Theorem 6.4 with k, S_k, S'_k as above, together with $l = 5_3$, $S_l = \{1, 5_2, 5_3, 7_1, 7_2\}$ and $S'_l = \{1, 5_2, 5_3\}$. We get a subsystem E_l obtained from E_k by $u_j \mapsto w_j$ and $(v_0, v_1, v_2, v_3, v_4, v_5, v_6) \mapsto (z_2, z_1, z_0, z_4, z_3, z_6, z_5)$, with $w_0 = x_l(5_2,5_2), w_1 = x_l(7_1,5_2), w_2 = x_l(7_2,5_2), z_0 = y_l(5_2,5_2), z_1 = y_l(5_3,5_2), z_2 = y_l(5_3,5_3), z_3 = y_l(7_1,5_2), z_4 = y_l(7_1,5_3), z_5 = y_l(7_2,5_2), z_6 = y_l(7_2,5_3)$. The common simple summands in Theorem 6.4 give the extra equation

$$\underbrace{1/125}_{i=1} + \underbrace{5u_0z_2}_{i=5_3} + \underbrace{7u_1z_4}_{i=7_1} + \underbrace{7u_2z_6}_{i=7_2} - u_0 = 0. \tag{7.1}$$

Let J be the ideal over $\mathbb{Q}$ generated by these two explicitly specified systems and the left-hand side of (7.1). We verify $1 \in J$ by the exact certificate described below, whose reduction relations and polynomial multipliers are supplied in the supplement.

After rational rescaling and three linear eliminations in each system, the certificate expresses every recorded reduction relation as an explicit polynomial combination of the remaining input equations. The leading monomials of these relations leave 14 monomials for each system, so their 196 pairwise products span the quotient for the two systems together. Reducing the product of each spanning monomial with the rescaled left-hand side H of (7.1) gives a 196×196 rational matrix M. All these reductions are defined over $\mathbb{Z}_{(101)}$, and the determinant of the resulting matrix modulo 101 is 85. Hence M is invertible over $\mathbb{Q}$. It follows that 1 is a multiple of H modulo the certified reduction relations, and therefore $1 \in J$. Linear independence of the spanning monomials is not needed for this implication. Thus the necessary equations have no common solution in characteristic zero.

The files `GUIDE.md` and `CERTIFICATE.md` identify the input equations and explain the certificate. The Python verifier checks the polynomial identities, the linear eliminations, the monomial reductions, and the determinant using exact arithmetic.

□

The character table provides another useful description of $\mathcal{F}_{210}$ and explains the role of its dimension characters. For a commutative fusion ring, the commuting normal fusion matrices can be simultaneously diagonalized. Their joint eigenvalue columns are precisely the unital ring homomorphisms to $\mathbb{C}$, called *linear characters*. This follows either from the regular representation of the commutative semisimple algebra $\mathcal{F} \otimes_{\mathbb{Z}} \mathbb{C}$, or directly from $M_i M_j = \sum_k N_{ij}^k M_k$.

Writing $\zeta_m = \exp(2\pi \mathrm{i}/m)$, the character table of $\mathcal{F}_{210}$, in the displayed basis order, is

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 5 & -1 & -\zeta_7-\zeta_7^6 & -\zeta_7^5-\zeta_7^2 & -\zeta_7^4-\zeta_7^3 & 0 & 0 \\ 5 & -1 & -\zeta_7^5-\zeta_7^2 & -\zeta_7^4-\zeta_7^3 & -\zeta_7-\zeta_7^6 & 0 & 0 \\ 5 & -1 & -\zeta_7^4-\zeta_7^3 & -\zeta_7-\zeta_7^6 & -\zeta_7^5-\zeta_7^2 & 0 & 0 \\ 6 & 0 & -1 & -1 & -1 & 1 & 1 \\ 7 & 1 & 0 & 0 & 0 & \zeta_5+\zeta_5^4 & \zeta_5^2+\zeta_5^3 \\ 7 & 1 & 0 & 0 & 0 & \zeta_5^2+\zeta_5^3 & \zeta_5+\zeta_5^4 \end{pmatrix}.$$

The formal codegree of a character is

$$n_\chi = \sum_i \chi(b_i)\chi(b_i^*). \tag{7.2}$$

These are the eigenvalues, counted with multiplicity, of multiplication by $c_{\mathcal{F}} = \sum_i b_i b_i^*$ on $\mathcal{F} \otimes_{\mathbb{Z}} \mathbb{C}$; see [30, §2.3]. A character is *nonvanishing* if it is nonzero on every basis element. The columns above have formal codegrees $210, 6, 7, 7, 7, 5, 5$, respectively. In particular, only the Frobenius–Perron column is nonzero on every basis element.

A commutative fusion ring is *dual-Burnside* if a linear character is nonvanishing on its basis exactly when its formal codegree equals the global Frobenius–Perron dimension [7, §12.2]. For a finite group G, columns of the ordinary character table are indexed by conjugacy classes, and the column of g has formal codegree $|C_G(g)|$. Thus the character ring of a centerless group is dual-Burnside precisely when every column except the identity column has a zero entry. Every non-abelian, non-alternating finite simple group has this property except M_{22} and M_{24}; see [7, §12.2].

Lemma 7.3. *A pivotal categorification over $\mathbb{C}$ of a commutative dual-Burnside fusion ring is pseudo-unitary.*

Proof. The pivotal dimension character is nonzero on every simple object. Its formal codegree is the global categorical dimension, so the dual-Burnside condition gives $\dim(\mathcal{C}) = \mathrm{FPdim}(\mathcal{C})$, which is pseudo-unitarity. Such a category admits a spherical structure with simple dimensions equal to their Frobenius–Perron dimensions [11, Proposition 9.5.1]. □

In fact, every commutative dual-Burnside fusion ring is weakly integral [7, Theorem 1.13 and the following discussion], so all its complex categorifications are pseudo-unitary [11, Proposition 9.6.5]. The displayed table shows that $\mathcal{F}_{210}$ is dual-Burnside. The Python verifier checks every character multiplication identity and all displayed formal codegrees using exact cyclotomic arithmetic. Although integrality already gives pseudo-unitarity in its characteristic-zero application, this observation is useful for other commutative fusion rings: it identifies pseudo-unitarity from the behavior of their nonvanishing characters. Over a field $\Bbbk$ of positive characteristic, a linear character means a unital homomorphism $\mathcal{F} \to \Bbbk$. The pivotal dimension character is still nonvanishing, and the formal codegrees lead to the following lifting test.

Theorem 7.4. *Let $\mathcal{F}$ be a commutative fusion ring and let $\Bbbk$ be an algebraically closed field of characteristic $p > 0$. If $\mathcal{F}$ has a pivotal categorification over $\Bbbk$, then it has a nonvanishing linear character $\chi : \mathcal{F} \to \Bbbk$. If, in addition, all its complex formal codegrees are integers prime to p, then $\mathcal{F}$ has a categorification over $\mathbb{C}$.*

Proof. The pivotal dimension is a ring homomorphism, and the dimension of every simple object is nonzero [11, Propositions 4.7.12 and 4.8.4]. It therefore gives the required nonvanishing character χ. Let M be the integral matrix of multiplication by $c_{\mathcal{F}}$. Its characteristic polynomial is $P(t) = \prod_j (t - n_j)$, where the n_j are the formal codegrees. Applying the dimension character to the Cayley–Hamilton identity gives

$$0 = P\big(\chi(c_{\mathcal{F}})\big), \qquad \chi(c_{\mathcal{F}}) = \sum_i \dim(X_i)\dim(X_i^*) = \dim(\mathcal{C}).$$

These equalities are taken in $\Bbbk$, so if all n_j are integers prime to p, then $\dim(\mathcal{C}) \neq 0$. The lifting theorem [11, Theorem 9.16.1] produces a characteristic-zero fusion category with the same fusion rules. Its finitely many structure constants descend to a finitely generated characteristic-zero field, which embeds into $\mathbb{C}$.

□

Corollary 7.5. *The fusion ring $\mathcal{F}_{210}$ admits no pivotal categorification in positive characteristic.*

Proof. By §6.3, any fusion categorification of these rules is split. Extending its morphism spaces to an algebraic closure and taking direct sums of the same simples preserves its fusion rules and pivotal structure. We may therefore take $\Bbbk$ algebraically closed. Multiplication by $\sum_i b_i b_i^*$ in $\mathcal{F}_{210}$ has characteristic polynomial

$$(t - 210)(t - 6)(t - 5)^2(t - 7)^3.$$

This follows by multiplying the displayed fusion matrices (see also the direct verification). Theorems 7.2 and 7.4 therefore exclude every characteristic outside $\{2,3,5,7\}$.

The remaining characteristics admit an elementary argument entirely inside the integral fusion ring. Write

$$R = 1 + 5b_{5_1} + 5b_{5_2} + 5b_{5_3} + 6b_{6_1} + 7b_{7_1} + 7b_{7_2}, \qquad B = b_{5_1}b_{5_2}b_{5_3}b_{6_1}b_{7_1}b_{7_2}.$$

The fusion rules give

$$B = 175R, \qquad R^2 = 210R, \qquad B^2 = 36750B. \tag{7.3}$$

For completeness, the coefficient vectors of the successive products defining B, in the displayed basis order, are

$b_{5_1}b_{5_2}$	0	0	1	0	1	1	1
$b_{5_1}b_{5_2}b_{5_3}$	0	3	3	3	4	4	4
$b_{5_1}b_{5_2}b_{5_3}b_{6_1}$	4	18	18	18	21	25	25
$b_{5_1}b_{5_2}b_{5_3}b_{6_1}b_{7_1}$	25	125	125	125	150	175	175
B	175	875	875	875	1050	1225	1225

The identity $R^2 = 210R$ also follows from $b_iR = d_iR$ and $\sum_i d_i^2 = 210$. If $p \in \{2,3,5,7\}$, then $36750 = 0$ in $\Bbbk$, so every character $\chi : \mathcal{F}_{210} \to \Bbbk$ satisfies $\chi(B)^2 = 0$ and hence $\chi(B) = 0$. Since B is the product of the six nonunit basis elements,

$$0 = \chi(B) = \chi(b_{5_1})\chi(b_{5_2})\chi(b_{5_3})\chi(b_{6_1})\chi(b_{7_1})\chi(b_{7_2}).$$

Since $\Bbbk$ is a field, some factor vanishes, contradicting the nonvanishing of the pivotal dimension character. □

Proposition 7.6. *For every field $\Bbbk$ of characteristic 2, 3, 5 or 7, the algebra $\mathcal{F}_{210} \otimes_{\mathbb{Z}} \Bbbk$ is not semisimple.*

Proof. The element R of (7.3) has unit coefficient 1, and $R^2 = 210R = 0$ in $\Bbbk$. Thus the commutative algebra has a nonzero nilpotent element. □

Question 7.7. *Does $\mathcal{F}_{210}$ have a non-pivotal fusion categorification in positive characteristic? Separately, can it occur as the Grothendieck ring of a non-semisimple finite tensor category in positive characteristic?*

Remark 7.8 (Further localizations and an alternative obstruction)**.** A further test case considered in [1, Question 1.14 and §6.5] is a rank-9 candidate of Frobenius–Perron dimension 504 with the same type as $\mathrm{Rep}(\mathrm{PSL}(2,8))$ but different fusion rules. Its multiplicity-free squares give larger localizations of the same kind. Another test case in the interpolated family is $q = 10$: the associated fusion ring has rank 11, dimension 990, and type $[[1,1],[9,5],[10,1],[11,4]]$. Several choices of repeated lateral label lead to overlapping subsystems. These examples indicate concrete directions for extending the $q = 6$ argument; checking the consistency and compatibility of their subsystems is a separate problem.

An alternative approach to $\mathcal{F}_{210}$, suggested by discussions with Scott Morrison and Pavel Etingof, is to seek an induction matrix for the adjoint of the forgetful functor $\mathcal{Z}(\mathcal{C}) \to \mathcal{C}$ (see [1, 28]). Proving that no such matrix exists would give a different obstruction. Such an approach requires controlling possible induction data; the localized triangular prism equations provide an alternative that does not enumerate those data.

7.2. **Fusion ring $\mathcal{F}_{660}$.** The simple integral fusion ring $\mathcal{F}_{660}$ has rank 8 and Frobenius–Perron dimension 660. The basis dimensions, in the order below, are $1, 5, 5, 10, 10, 11, 12, 12$. Its fusion matrices are

$$\begin{smallmatrix}
1&0&0&0&0&0&0&0\\0&1&0&0&0&0&0&0\\0&0&1&0&0&0&0&0\\0&0&0&1&0&0&0&0\\0&0&0&0&1&0&0&0\\0&0&0&0&0&1&0&0\\0&0&0&0&0&0&1&0\\0&0&0&0&0&0&0&1
\end{smallmatrix},\quad
\begin{smallmatrix}
0&1&0&0&0&0&0&0\\0&0&1&1&1&0&0&0\\1&0&0&0&0&0&1&1\\0&0&1&0&1&1&1&1\\0&1&0&0&1&1&1&1\\0&0&1&1&0&1&1&1\\0&0&0&1&1&1&1&1\\0&1&0&1&1&1&1&1
\end{smallmatrix},\quad
\begin{smallmatrix}
0&0&1&0&0&0&0&0\\1&0&0&0&0&0&1&1\\0&1&0&1&1&0&0&0\\0&1&0&0&1&1&1&1\\0&1&0&0&1&1&1&1\\0&0&1&1&1&1&2&2\\0&0&1&1&1&1&2&2\\0&0&1&1&1&1&2&2
\end{smallmatrix},\quad
\begin{smallmatrix}
0&0&0&1&0&0&0&0\\0&0&1&1&1&0&0&0\\0&1&0&1&0&1&1&1\\0&0&1&1&0&1&1&1\\1&0&0&3&1&1&2&2\\0&1&1&1&1&2&2&2\\0&1&1&1&1&2&2&2\\0&1&1&1&1&2&2&2
\end{smallmatrix},\quad
\begin{smallmatrix}
0&0&0&0&1&0&0&0\\0&0&1&1&1&0&0&0\\0&0&0&1&1&1&1&1\\0&1&0&1&1&1&1&1\\0&1&1&2&1&2&2&2\\0&1&1&2&1&2&2&2\\0&1&1&2&1&2&2&2\\0&1&1&2&1&2&2&2
\end{smallmatrix},\quad
\begin{smallmatrix}
0&0&0&0&0&1&0&0\\0&0&1&1&1&0&0&0\\0&0&0&1&1&1&1&1\\0&1&0&1&1&1&1&1\\1&1&1&2&2&2&2&2\\0&1&1&2&2&2&2&2\\0&1&1&2&2&2&3&2\\1&1&1&2&2&3&3&3
\end{smallmatrix},\quad
\begin{smallmatrix}
0&0&0&0&0&0&1&0\\0&0&1&1&1&0&0&0\\0&0&0&1&1&1&1&1\\0&1&0&1&1&1&1&1\\0&1&1&2&2&2&2&2\\0&1&1&2&2&2&3&2\\0&1&1&2&2&2&3&3\\0&1&1&2&2&3&3&3
\end{smallmatrix},\quad
\begin{smallmatrix}
0&0&0&0&0&0&0&1\\0&0&1&1&1&0&0&0\\0&0&0&1&1&1&1&1\\0&1&0&1&1&1&1&1\\0&1&1&2&2&2&2&2\\0&1&1&2&2&2&3&2\\0&1&1&2&2&3&3&3\\1&1&1&2&2&3&3&2
\end{smallmatrix}$$

Number these basis elements $b_0, \dots, b_7$ in the displayed order, with $b_0 = 1$, and write N_{ab}^c for their structure constants. The duality involution is (1 2), fixing all other indices. The matrices use the convention $(M_a)_{bc} = N_{ab}^c$.

Theorem 7.9. *The fusion ring $\mathcal{F}_{660}$ has no fusion categorification over any field.*

Proof. Apply Theorem 6.7 to the zero-based tuple

$$(i_1, i_2, i_3, i_4, i_5, i_6, i_7, i_8, i_9) = (1, 3, 4, 1, 1, 3, 4, 2, 2).$$

The six required fusion coefficients, in the order used in that theorem, are

$$(N_{11}^3, N_{11}^3, N_{13}^4, N_{42}^1, N_{34}^2, N_{22}^4) = (1, 1, 1, 1, 1, 1), \qquad N_{31}^4 = 1.$$

The following table lists all coefficients needed for the empty-spectrum condition and the two one-dimensionality conditions:

k	0	1	2	3	4	5	6	7
N_{14}^k	0	0	1	1	0	1	1	1
N_{22}^k	0	1	0	1	1	0	0	0
N_{31}^k	0	0	1	0	1	1	1	1
N_{11}^k	0	0	1	1	1	0	0	0
N_{42}^k	0	1	0	1	0	1	1	1

The first three rows have no position at which all entries are nonzero. Consequently

$$\sum_k N_{14}^k N_{22}^k N_{31}^k = 0.$$

The last two rows have a unique common nonzero position, $k = 3$, and the second and third rows have a unique common nonzero position, $k = 4$. Thus

$$\sum_k N_{11}^k N_{42}^k = 1, \qquad \sum_k N_{31}^k N_{22}^k = 1.$$

These are respectively the second alternative in (6.11) and the first alternative in (6.12). All hypotheses of the zero-spectrum criterion hold as integer identities. Theorem 6.7 therefore applies over every field. □

The proof uses semisimplicity when it resolves tensor products into simples. The existence of a nonsemisimple finite tensor category with Grothendieck ring $\mathcal{F}_{660}$ is a separate question.

Data availability. The computational supplement is available at `https://github.com/sebastienpalcoux/triangular-prism-categorification`. The version accompanying this article is commit `f816c69`. The proof of the minimum-rank assertion in Proposition 6.10 and its independent finite check are in `SPECTRUM_RANK.md` and `verify_spectrum_rank.py`. The reader's guide, `GUIDE.md`, explains how to supply another fusion ring, apply the zero- and one-spectrum criteria, and generate localization systems with explicit or symbolic categorical dimensions and the required additional data. The general entry point is `criteria.py`. The exact characteristic-zero certificate for $\mathcal{F}_{210}$ is explained in `CERTIFICATE.md` and checked by `verify_f210.py`. The entry point `verify.py` checks the worked applications, including the fusion tensors, character table, positive-characteristic identities, index tuples for the spectrum criteria, and stated census exclusions. Verification and equation generation use Python's standard library. The guide also describes optional certificate and group-data regeneration.

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References

[1] M. A. Alekseyev, W. Bruns, J. Dong and S. Palcoux, *Classifying integral Grothendieck rings up to rank 5 and beyond*, Lett. Math. Phys. **116** (2026), article 141, https://doi.org/10.1007/s11005-026-02164-7.
[2] J. W. Barrett and B. W. Westbury, *Invariants of piecewise-linear 3-manifolds*, Trans. Amer. Math. Soc. **348** (1996), no. 10, 3997–4022, https://doi.org/10.1090/S0002-9947-96-01660-1.
[3] B. Bartlett, *Fusion categories via string diagrams*. Commun. Contemp. Math. **18** (2016), no. 5, 1550080, 39 pp.
[4] P. Bonderson, *Non-abelian anyons and interferometry*. Caltech Ph.D. thesis (2007).
[5] N. Bourbaki, *Algèbre, chapitre 9: Formes sesquilinéaires et formes quadratiques*, Springer, Berlin, 2007, reprint of the 1959 edition, https://doi.org/10.1007/978-3-540-35339-3.
[6] W. Bruns and S. Palcoux, 1-*Frobenius simple integral fusion rings*, August 2025, census tables and fusion data.
[7] S. Burciu and S. Palcoux, *Burnside-type results for fusion categories*, Commun. Contemp. Math., article 2650007 (published online 2025), https://doi.org/10.1142/S0219199726500070.
[8] O. Davidovich, T. Hagge, and Z. Wang, *On arithmetic modular categories*, arXiv:1305.2229 (2013).
[9] F. Digne and J. Michel, *Representations of finite groups of Lie type*, second edition, London Math. Soc. Student Texts **95**, Cambridge University Press, Cambridge, 2020.
[10] J. Dong, S. Palcoux and A. Plessis, *Classification of vanishing fusion rings of rank four*, arXiv:2610.22000 (2026).
[11] P. Etingof, S. Gelaki, D. Nikshych, and V. Ostrik, *Tensor Categories*, Mathematical Surveys and Monographs **205**, Amer. Math. Soc., Providence, RI, 2015, https://doi.org/10.1090/surv/205.
[12] P. Etingof, D. Nikshych, and V. Ostrik, *On fusion categories*, Ann. of Math. (2) **162** (2005), no. 2, 581–642.
[13] P. Etingof, D. Nikshych, and V. Ostrik, *Weakly group-theoretical and solvable fusion categories*, Adv. Math. **226** (2011), no. 1, 176–205.
[14] J. Fuchs, I. Runkel and C. Schweigert, *A reason for fusion rules to be even*, J. Phys. A **35** (2002), no. 19, L255–L259, https://doi.org/10.1088/0305-4470/35/19/102.
[15] W. Fulton and J. Harris, *Representation theory. A first course*, Graduate Texts in Mathematics, 129. Readings in Mathematics. Springer-Verlag, New York, 1991. xvi+551 pp.
[16] H. He, Z. Liu, F. Lu, S. Palcoux and Y. Ren, *Triangular prism equations for near-group categories: algebraic solutions and a unitary conjecture*, arXiv:2610.11652 (2026).
[17] L. Huang, Z. Liu, S. Palcoux and J. Wu, *Complete positivity of comultiplication and primary criteria for unitary categorification*, Int. Math. Res. Not. IMRN 2024, no. 1, 817–860.
[18] A. M. Jaffe and Z. Liu, *Mathematical picture language program*, Proc. Natl. Acad. Sci. USA **115** (2018), no. 1, 81–86.
[19] I. Kaplansky, *Bialgebras*. Lecture Notes in Mathematics. Department of Mathematics, University of Chicago (1975) iv+57 pp.
[20] Z. Liu, *Exchange relation planar algebras of small rank*, Trans. Amer. Math. Soc. **368** (2016), no. 12, 8303–8348.
[21] Z. Liu, S. Palcoux and Y. Ren, *Classification of Grothendieck rings of complex fusion categories of multiplicity one up to rank six*. Lett. Math. Phys. **112** (2022), no. 3, Paper No. 54, 37 pp.

[22] Z. Liu, S. Palcoux and Y. Ren, *Interpolated family of non-group-like simple integral fusion rings of Lie type*, Internat. J. Math. **34** (2023), no. 6, Paper No. 2350030, 51 pp. https://doi.org/10.1142/S0129167X23500301.
[23] Z. Liu, S. Palcoux and J. Wu, *Fusion Bialgebras and Fourier Analysis: Analytic obstructions for unitary categorification*, Adv. Math. **390** (2021), 107905.
[24] G. Lusztig, *Leading coefficients of character values of Hecke algebras*, The Arcata Conference on Representations of Finite Groups, Proc. Sympos. Pure Math. **47**, Part 2, Amer. Math. Soc., Providence, RI, 1987, 235–262.
[25] S. Mac Lane, *Categories for the working mathematician*, second edition, Graduate Texts in Math. **5**, Springer-Verlag, New York, 1998.
[26] G. Mason, *A brief history of the positivity conjecture in tensor category theory*. Bull. Inst. Math. Acad. Sin. (N.S.) 14 (2019), no. 2, 149–153.
[27] J. Miquel Martínez, *An infinite family of counterexamples to a conjecture on positivity*, Rend. Sem. Mat. Univ. Padova **145** (2021), 205–206. https://doi.org/10.4171/RSMUP/79.
[28] S. Morrison and K. Walker, *The center of the extended Haagerup subfactor has 22 simple objects*, Internat. J. Math. **28** (2017), no. 1, article 1750009, 11 pp.
[29] S.-H. Ng and P. Schauenburg, *Higher Frobenius–Schur indicators for pivotal categories*, Hopf Algebras and Generalizations, Contemp. Math. **441**, Amer. Math. Soc., Providence, RI, 2007, 63–90, https://doi.org/10.1090/conm/441/08500.
[30] V. Ostrik, *Pivotal fusion categories of rank 3*. Mosc. Math. J. **15** (2015), no. 2, 373–396.
[31] D. Radchenko and C. Wheeler, *Real quadratic fields and finite quantum dilogarithms I*, arXiv:2609.21892v2 (2026).
[32] D. P. Thurston, *The algebra of knotted trivalent graphs and Turaev's shadow world*. Invariants of knots and 3-manifolds (Kyoto, 2001), 337–362, Geom. Topol. Monogr., 4, Geom. Topol. Publ., Coventry, 2002.
[33] V. G. Turaev and O. Ya. Viro, *State sum invariants of* 3*-manifolds and quantum* 6*j-symbols*, Topology **31** (1992), no. 4, 865–902.
[34] G. Vercleyen, *On Low-Rank Multiplicity-Free Fusion Categories*, PhD thesis, Maynooth University, 2024; also available as arXiv:2405.20075.
[35] G. Vercleyen and J. K. Slingerland, *AnyonWiki*, https://anyonwiki.org/, archived software version 1.0.2, https://doi.org/10.5281/zenodo.15863113.
[36] G. Vercleyen and J. K. Slingerland, *On low rank fusion rings*, J. Math. Phys. **64** (2023), no. 9, article 091703, 25 pp.
[37] G. Vercleyen and J. Slingerland, *On low-rank multiplicity-free complex fusion categories*, Commun. Math. Phys. **407** (2026), article 231, https://doi.org/10.1007/s00220-026-05705-2.
[38] Y. Wang, *On the positivity conjecture: a digest of Mason's counterexample*, unpublished note, https://yilongwang11.github.io/notes/On-the-positivity-conjecture.pdf.
[39] Z. Wang, *Topological quantum computation*, CBMS Regional Conference Series in Math. **112**, Amer. Math. Soc., Providence, RI, 2010.

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